Engineering Mechanics 1 · Unit_01 · UNIT 1

DRAFT — UE26ME142A · Unit_01 — UNIT 1 · Complete Unit Master

Notes-first study master · source-cited lecture explanations and compact page takeaways

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Read the edited lesson notes by topic. Evidence chips point to transcript intervals and slide pages. The complete source transcripts, page text, formula candidates, and image locators are preserved in sibling JSON datasets. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

Contents

1. 1 · UE26ME142A · Unit_01: UNIT 1
2. 2 · UE26ME142A_Unit 1_Class 1.1_AVSummary
3. 3 · UE26ME142A_Unit 1_Class 1_AVSummary
4. 4 · UE26ME142A_Unit 1_Class 2.0_AVSummary
5. 5 · UE26ME142A_Unit 1_Class 2.1_AVSummary
6. 6 · UE26ME142A_Unit 1_Class 2.2_AVSummary
7. 7 · UE26ME142A_Unit 1_Class 3_AVSummary
8. 8 · UE26ME142A_Unit 1_Class 4_AVSummary
9. 9 · UE26ME142A_Unit 1_Class 5_AVSummary
10. 10 · UE26ME142A_Unit 1_Class 6_AVSummary
11. 11 · UE26ME142A_Unit 1_Class 8_AVSummary
12. 12 · UE26ME142A_Unit 1_Class 9_AVSummary
13. 13 · UE26ME142A_Unit 1_Class 10.0_AVSummary
14. 14 · UE26ME142A_Unit 1_Class 10.1_AVSummary
15. 15 · UE26ME142A_Unit 1_Class 11_AVSummary
16. 16 · UE26ME142A_Unit 1_Class 12.0_AVSummary
17. 17 · UE26ME142A_Unit 1_Class 12.1_AVSummary
18. 18 · Source slide deck: UE26ME142A Unit 1 Class 1.0 Lecture PPT
19. Appendix A · Symbol glossary
20. Appendix B · Sources, coverage and gaps
21. Appendix C · Revision and retrieval practice

M1

Unit_01 · UNIT 1

1 UE26ME142A · Unit_01: UNIT 1

This unit master organizes source material for Engineering Mechanics 1, Unit_01, officially titled UNIT 1. It includes 16 video transcripts and 16 source slide decks, with page-level text and rendered images where available. This draft preserves source wording and page order; it does not claim a complete human subject review or full audiovisual review. EV0000001

SHORT-NOTE RECAP

Unit map · UE26ME142A · Unit_01: UNIT 1

Official catalogue entry: UE26ME142A, Unit_01 (UNIT 1). Bucket folder present: yes. Videos: 16; slide decks: 16.

EV0000001

2 UE26ME142A_Unit 1_Class 1.1_AVSummary

Study focus: UE26ME142A_Unit 1_Class 1.1_AVSummary. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000002 · EV0000003

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

This is class number two.

Key details

- I am Dr.
- Sv. from Civil Engineering under Department of Science and Humanities.
- under PS University.
- In the very first class we discussed a few things regarding this subject, that is what is indicated here in this slide.

- We came to know what is meant by mechanics, the different between statics and dynamics.
- Then the definitions, few of the definitions of space, mass, time, particle, rigid body, elastic body, plastic body.
- Then we define what is a force, what are the types of forces, what are the units of forces in S side.
- What are Newton's laws?
- And then we discussed about the characteristics of forces.
- In between, I also discussed about the prerequisite for understanding these subjects.
- That is mathematics, then physics, and of course.
- English that was required.
- This is class number two.
- First of the foremost thing is similar triangles under geometry.
- You all know what are similar triangles.
- One side and two angles or two sides and one angle, if they are proportional to each other in these two triangles, then we call them as similar triangles.
- If you take a triangle, any triangle, then.
- If a side is the distance of the side is taken as capital A, the angle opposite to this side in the triangle is taken as small A.
- Small b, the last side, if the distance is taken as capital C and the angle opposite to this side is taken as small a, then the sides of the triangle.
- Will be proportional to the sine of the angle opposite to the other two angles or opposite to this side or angle between the other two sides.
- So that means A by capital A divided by sine of small A, that is the angle made by the.
- other two sides other than capital A is small.
- What is law of cosine?
- Again, we take a triangle, any general triangle.
- If the sites are named as the distances of the sites are named as A, B, and C, and the angle opposite to side A is taken as small A.
- Then we can say that the distance of the side A, capital A, A square is equal to.
- B square plus C square minus 2 B C into cos of angles A.
- So that means.
- When you have a triangle, we can determine the any side of the triangle if we know two sides and one angle which is opposite to the side of the distance for which we have to calculate, then we can do it easily by using law of cosines.
- Next comes what is complementary angle.
- All of that complementary angles sum up to 90 degrees.
- Supplementary angle is the sum of two angles will come to 180 degrees.
- Corresponding angles will always be equal.

Example or application from the lecture

- Then let's understand what is law of science.
- Next, let us look at what is law of cosine.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000002 · EV0000003 · EV0000004 · +1 source units in pack.json

Core explanation · 05:00–09:59

Alternate angles are also will have equal angles between them.

Key details

- The vertically opposite angles also will be equal to each other.
- And of course tripometry basics of trigonometry is required.
- Then of course basics of integration is also required to certain extent.
- Next comes this.
- Few things that we need to understand.
- That is first one is deformation.
- What is deformation?
- No, when you take a body and there is a force applied onto this body, the body's shape will change.
- But the position of the body would not change.
- Changes when you apply a force is called as deformation.
- The position of the body does not change, only the shape of the body changes.
- Will change whereas the position does not change.
- The next one is displacement.
- This change in the position of the object because of the application of the force is called as displacement.
- The body is getting displaced from one point to another point because of application of the force.
- What is defluction?
- We have two poles and in between we have an electrical wire.
- Over a period of time, because of its own weight, this wire would actually bent or it will take a shape of a bow okay when you have a bow inverted or kept it horizontal you can see that it will have a shape like the sector of a s curve okay or it will take the shape of a curve it's a sector of a circle okay.
- Now that is because of the sulfate of the wire itself.
- No, if you take any point on this wire, the.
- Body has undone the shape change because it was horizontal, now it has changed to a curve one.
- And again the position of that point would have changed, it would have come down.
- So there is both deformation and displacement occurring simultaneously in an object, such changes is defined as deflection.
- There could be a change in the position of the body, or the shape of the body can change, or there could be deflection.
- Any of this or combination of this can occur.
- In the last class also, I talked about coordinate system, whether it is three-dimensional or two-dimensional.

Example or application from the lecture

- The best example is you take a sponge and when you apply force onto this, the shape of the sponge.
- That's the best example for understanding what is deformation.
- Now, you take a body, and when you apply a force onto body, for example, let me take this pen.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000006 · EV0000007 · EV0000008 · +1 source units in pack.json

Core explanation · 10:00–14:59

But I also told you that we use the right hand first three fingers to take or to note the directions of the coordinate system.

Key details

- Then the forefinger will be y-axis and the middle finger will be the z axis.
- The three fingers that I am taking of my right hand is also mutually perpendicular to each other.
- To each other.
- But when I if I apply a force onto this, say I will take this as x-axis and this as y-axis, and I am applying a force onto this pen along the x-axis or only along the y-axis or only in the plane x y.
- I'm not going to be applying the force normal to this XY or along the Z axis, then I can represent these forces only in these two planes, that is X and Y.
- Forces onto this body will also occur in the same plane.
- Then we can treat this as a two-dimensional force system.
- Next, we will look at what is translatory motion.
- I talked about displacement earlier when I told you that if there is a force applied on to a body, if the position of the body changes from one position to another position because of application of the force.
- that without any change in the shape of the body, then we call that as translatory motion.
- The translatory motion can occur along x-axis if I take if I take this, okay.
- If the body moves only along X, that is translatory along X.
- If the body moves only along Y, it is Y translatory motion.
- The body can move along Z also.
- Forward and backward with respect to the screen, okay.
- Z axis, that is also translatory motion.
- There could also be rotatory motion because of the application of the force.
- What is this rotatory motion?
- Because of which, and I take this as x axis, the thumb is x-axis, the forefinger is the y-axis, and the.
- Middle finger is an axis.
- If on application of the force, if this pen rotates, rotates about the x-axis, this is x axis, okay, this is x axis, if it rotates about the x axis like this, okay.

- Then that is translatory motion along X.
- Y axis, this is the Y axis, if the body rotates along the y-axis, then it is rotation along y.
- Then you have z axis here, if the body rotates along the z axis, then it is rotation along the z axis.
- Any translatory motion of this point because of application of the force can be represented along this three-coordinate system that is translatory motion along X, translatory motion along Y, and translatory motion along Z.
- Best thing that we can do is or to understand this is you take a plane.
- A plane which is moving.
- If it is moving along X direction, then that is a transfer to motion along X.
- Then if it rotates like this, this is rotation important.

Example or application from the lecture

- For example, if I take the thumb as x-axis.
- Now, if I take an object, say for example I take this pen, okay, this pen has got length and it has got diameter, so it's a three dimensional body.
- The coordinate system that I can consider can be only 2D coordinate system.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000010 · EV0000011 · EV0000012 · +1 source units in pack.json

Core explanation · 15:00–19:59

The x-axis, okay.

Key details

- Then if the plane goes up and down, up and down, that is translatory motion along y.
- If the plane rotates along the horizontal plane, if it rotates like this, if it rotates like this, then this is rotation along y.
- Then if the plane moves perpendicular to the screen, to and fro, okay, that is transmittery motion along Z, and if it rotates like this, then it is rotation along Z axis.
- I hope all of you have understood what are the different coordinate systems and what is translatory motion and rotatory motion.
- Then if I take the x axis that is horizontal axis and the vertical axis as y, so the plane is perpendicular to our screen.
- It can move along Y?
- It cannot move along the Z because once we say it is moving along Z, then it is it is coming out of the plane.
- Say I take this pen itself, which is in this plane, okay?
- If it has to rotate about x, then it has to come out of the plane.
- Can it rotate about the z axis?
- No, this is the rotation along about z axis.
- No, it is not possible, it is coming out of the plane, no.
- But can it rotate about the z axis?

- Two transfer motion, one along X and one along Y, if I am taking XY plane, and it can rotate about the Z axis.
- It cannot rotate about the X axis, it cannot rotate about the Y axis.
- Of course in addition to this mathematics, all of you should be knowing about addition, subtraction,.
- Division and multiplication.
- Along with this, you should also know what is dot product when it comes to vectors because all forces are vectors.
- vector and cross product of vector.
- These things have to be known by you before we start to understand engineering mechanics statics.
- Close today's discussion.
- This is class number two on engineering mechanics statics, which is a subject which is common to all students who are entering B Tech program in the first year.
- I am Dr.
- SV Venkatesh, a civil engineering.
- from Department of Science and Humanities under PS University.
- Hope to see you in the next class.
- That is class number three.

Example or application from the lecture

- Now, if we are going to be considering only two-dimensional, okay, say for example, I'm going to take a plane which is perpendicular to the screen, okay.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000014 · EV0000015 · EV0000016 · +1 source units in pack.json

3 UE26ME142A_Unit 1_Class 1_AVSummary

Study focus: UE26ME142A_Unit 1_Class 1_AVSummary. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000018 · EV0000019

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

Dear student, welcome all of you for engineering mechanics classes.

Key details

- I am Dr.
- Svinkitesh, civil engineering from the Department of Science and Humanities.
- I welcome all of you for the first year courses under PS University.
- I will be teaching you the subject engineering mechanics statics.

- which is a common subject for all students who are entering into B Tech program under PS University.
- I will start with the first chapter, which is Introduction to Statics.
- For this subject, we are using a textbook by the title Engineering Mechanics Statics.
- And we are using the eighth edition.
- This addition is in SI version.
- for this book is JL Meriam and LG Craig.
- And you can see here the front covering page or the face page of the book.
- It comes under two different colours, but the contents are same.
- The eighth edition is adopted by.
- J and Bolton and that's what you can see here.
- The author's name is LJ Craig.
- Sorry, J L Miriam, L G Craig, J and Bolton, and it's the same thing here also.
- But only the difference between these two is the colour.
- One is in red colour, the other one is in green colour.
- You can buy whichever is available in the market.
- Both of them, the content inside is same, so you can take up whichever book that is available in the market.
- So in the first introduction chapter, we will be dealing with this.
- First thing is what is mechanics?
- Mechanics is that branch of engineering which deals with the study of bodies which are under rest.
- Which are in motion.
- There are two aspects for this.
- The first one is the bodies which are at rest, which is called as statics.
- The bodies which are in motion.
- comes under dynamics.
- One is the steady of bodies which are at rest which is called statics.
- The other one is the bodies which are in motion which is comes under dynamics.
- Few of the basic definition that we need to know is listed here.
- Of course, all this you would have gone through in your earlier classes, which is before joining your MTEC, you would have gone through this.
- And all of you have studied in English medium, or you would have changed from some other medium to English medium.
- Then of course physics.
- This subject is actually an extension of whatever you have learned in your physics.
- There, you would have learned the basics.
- Here you are going to learn the application of the basics with respect to engineering.
- Then you should know mathematics.
- In mathematics, all the basics that you have learned is what is required.
- Then algebra, linear algebra.
- Then trigonometry, then integration, which is calculus.
- And of course very importantly geometry.
- The geometry plays a very important role when you have to solve some problems.

- As we go on, you will come to know what are the things that we should know.

Example or application from the lecture

- For example, arithmetic, which is addition, subtraction, multiplication, and division must ensure.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000018 · EV0000019 · EV0000020 · +1 source units in pack.json

Core explanation · 05:00–09:59

All of you would have known what is space.

Key details

- How do you define space?
- You should know what is space.
- Space is the object.
- which occupies the point which can be measured with respect to the three coordinate systems.
- We have three coordinate systems for defining any point on this earth or in surrounding.
- We need to, we or we ref we refer to three coordinate systems that is x-axis, y axis, and the z axis.
- To know this, we use the right hand first three fingers, that is the thumb, the four finger, and the middle finger, and this are mutually perpendicular to each other.
- If you take the thumb, the thumb if you take it as the x-axis, the forefinger represents the y axis and the middle finger represents the z axis.
- By the three coordinate systems, the distance as X, Y, and Z, which talks about what is the X distance of that point with from the origin measure along the x-axis.
- Then y distance is what is the distance measured from the origin.
- Along the y axis, then Z distance is the distance measured from the origin along the z axis.
- This is one way of identifying a point in space with respect to the three different axes.
- We can also locate.
- The point by using the angular measurements.
- That is what is the angle made by this point in space with respect to the x axis that is theta x, then with respect to y axis that is theta y, then with respect to the z axis that is theta z.
- Either we can have this three distance measurement or three angular measurement or the combination of this distance measurement or the angular measurement.
- Between this is 3D.
- 3D is when we use all the three coordinate systems, we go for what is called as three-dimensional objects.
- Then we can define that or we can represent it as a two-dimensional object.
- 3D is when we have to define.
- The object by all the three coordinate systems, then we go for 3D, or it is called as three dimensional objects.
- The next in the definition is mass.

- What is mass?
- It is the amount of substance that is present in a body is called as mass.
- Then how do we define time?
- Time is defined as the measure of succession of events.
- The instant that we leave our house and the instant that we reach the college, that measure between the instant that we leave our house and the instant that we reach the college, that measure of succession of events is called as time.
- Next, how do we define a particle?

Example or application from the lecture

- If we can represent the object and the forces acting on them along any one plane using only two coordinate systems, for example, only X and Y.
- For example, if we have to measure the or we have to know the time elapsed between.
- We don't consider time when we are talking about objects which are at rest, that is, when we are dealing with statics.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000022 · EV0000023 · EV0000024 · +1 source units in pack.json

Core explanation · 10:00–14:59

A particle is defined as an object which occupies a single point in space which has mass, which has negligible dimension.

Key details

- With respect to naked eyes.
- Of course, when you go to nano level, you can certainly measure the dimensions of a particle.
- Point which is occupied by the object in space, one single point.
- That means we can measure it with respect to the coordinate system X, Y, and Z.
- Then when we say it has mass, the particle will settle down.
- Then it is very tiny and we cannot measure it with the naked eyes.
- Such objects are defined as particles.
- Next, the three different types of bodies.
- What is a rigid body?
- A rigid body is one in which when we apply force onto this, there is no change in shape and size of the body.
- Such bodies are defined as rigid body.
- Next, elastic body.
- What is an elastic body?
- When a force is applied on an elastic body, it changes its shape and size.
- But on removal of the force causing this change, the body will regain its original shape and size.
- Such bodies are called as elastic bodies.
- Next is a plastic body.
- What is a plastic body?

- A plastic body is one.
- Which will change its shape and size on application of a force, but it will not regain its original shape and size.
- Such bodies are called as plastic bodies.
- Next, how do we define a force?
- A force is something which changes the state of rest of uniform motion of an object.
- If the object or the body is already in motion, it can stop that motion or it can increase the speed of the body or it can reduce the speed of the body.
- For such application or such thing is called as a force.
- Now, next comes what are the different types of forces.
- Forces can be broadly divided into two types.
- Forces which are having forces which are in one plane, coplanar forces.
- The other one is non-coplanar forces.
- Which can be represented only along two dimensions.
- Non-co-planar forces are forces which can be represented along all the three coordinate systems that is required.
- Concurrent forces and non-concurrent forces.
- What are concurrent forces?
- Concurrent forces are forces which will have the same point of application.
- Non-concurrent forces which will have different points of application.
- Under non-concurrent force system, we can have parallel force system and non-parallel force system.
- Under parallel force system, all the forces are going to be parallel to each other.
- Non-parallel force system, they need not have to be parallel.
- Under parallel force system, we can have two categories.
- One is like parallel force system and unlike parallel force system.
- What is like parallel force system?
- Like parallel force system, where the sense of the force are same for all.
- Unlike parallel force system, the sense of the forces are different.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000026 · EV0000027 · EV0000028 · +1 source units in pack.json

Core explanation · 15:00–19:59

Next coming to what is the you what are the units of forces.

Key details

- Unit of forces in SI, that is system of interaction.
- SI is Newtons.
- And you can see the picture of Sir Isaac Newton.
- Which is pulled by the gravitational force.
- 9.81 newtons means it is one kg of force which is pulled by a gravitational force that is 9.81.
- 1 into 9.81 meters square meter per second square.
- What are Newton's laws?

- There are three Newton laws.
- All of you have studied this.
- Newton's law one, two, and three.
- What is Newton's law one?
- Newton Law 1 states that.
- Any body would continue to be in its state of rest or of uniform motion unless and until an external force is applied onto it.
- Is what you have studied.
- But here, instead of saying external force, we say that unless and until it is.
- It is unbalanced force that is acting.
- Where is that force coming?
- It is because the internal that is developed by the person.
- A body will continue to be in its state of rest or of uniform motion unless and until there is an unbalanced force acting on it.
- The second law states that the force is proportional to the mass of the body.
- Acceleration is the proportionality constant.
- F is equal to M.
- Of course, the Newton's third law, all of every action has an equal and opposite reaction.
- Which you already know or studied.
- That is what are the characteristics of forces.
- What is the characteristics of force?
- Force one is the magnitude.
- Magnitude is defined as the amount of force that is acting.
- The amount of quantity of the force is called as magnitude.
- Next is the direction.
- The direction is the angle made by the force with respect to a reference axis.
- The reference axis can be anything.
- The third one is the.
- Point of application.
- Where are we supposed to apply this?
- That is the point of application.
- The fourth one is the sense of the force.
- Whether it is acting away from the point or towards the point.
- In the book, it clubs the direction and the sense together.
- Garney says that line of action of the force.
- Then your direction if it is theta and if it is moving towards the angle will become 180 plus theta.
- So this much we can understand in the first class.
- We will continue this in the next class.
- And I am Dr.
- Svang Tesh, civil engineering.
- from the Department of Science and Humanities, PS University.
- See you in my next class where we will be discussing further into the introduction of statics.

Example or application from the lecture

- For example, a person who is standing, he can walk.
- Let's look at this.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000030 · EV0000031 · EV0000032 · +1 source units in pack.json

4 UE26ME142A_Unit 1_Class 2.0_AVSummary

Study focus: UE26ME142A_Unit 1_Class 2.0_AVSummary. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000034 · EV0000035

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

My name is Dr.

Key details

- Swivin Kitish from Civil Engineering under Department of Science and Humanities, PS University.
- This is class number five.
- In the last class we discussed about the dot product, we also looked about how to use the law of parallel of forces and to get the resultant.
- Now that is also a resolution, but in general, resolution of a force means splitting up of a force into N number of forces such that N number of forces would produce the same effect as that of a single force.
- That means you have one force, it will have some effect on the body.
- It can be translatory motion or it can be rotatory motion or it can be both translatory as well as rotatory.
- Next look at moment.
- What is moving a force?
- Earlier I had discussed about cross product.
- Cross product is the position vector cross the force vector.
- It is not a force vector cross the position vector.
- That means the distributory property is not valid in case of finding out the cross product.
- It should always be the position vector cross the force vector.
- The magnitude, whatever we get, is nothing but moment.
- Now there are two ways of getting this moment, or first I will define what is the definition of moment.
- The rotating effect produced by a force on a body is called as moment.
- Force into perpendicular distance.
- When I say perpendicular distance, the distance between the point about which we want the movement and the point about which the force is acting such that the force is acting at 90

degrees between the point where the movement is required.
- On the point where the force is acting, or we find out what is the perpendicular distance between these two.
- The cable, okay, when we apply external forces such that it is moving away from each other, the body would try to elongate.
- Now this elongation that happens because of the external forces that is applied is called as tension.
- towards each other.
- The internal forces when it acting towards each other, the body won't get elongated.
- When we look at from this end, how do we look at this is point A, where the cable is fixed to this clamp.
- The cable is indicated by an arrow here.
- The force, that force is moving away from point A and on the other side, if it is point B, that would also be moving away.
- That is the internal force.
- How where do we get this tension?
- We have this cable.
- Which is used to carry some load here.
- This load is put up on a platform.
- There is some load, and this entire load with the platform is lifted by the crane.

Example or application from the lecture

- As well, we looked at how do we resolve the given force along XY direction, determine their components, and then how do we use this component to determine the result.
- Next look let us look at what do we mean by tension.
- A cable that is given here, which is attached to this.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000034 · EV0000035 · EV0000036 · +1 source units in pack.json

Core explanation · 05:00–09:59

Upwards, whereas here, because of this weight, this will be pulled downwards.

Key details

- towards each other so the cable will experience tensile force whereas if you see the external upwards it is being pulled upwards because of the weight it is not being pulled downwards so external forces is acting away from each other like this.
- But the vice versa if you see compression.
- When you apply forces which is moving, the external forces which is moving towards each other, the body would try to get compressed.
- Next, we will define what is called as principle of transmissibility of forces.
- O is an hint support.
- We will come back to this type of support later when we take up an equilibrium chapter.
- And at C we have a roller support.

- This body is.
- Force P at B point.
- It states that.
- The point of application of a force can be along the line of action, anywhere along the line of action of the force.
- Such that it produces the same effect on the body, provided the point is in contact with the body.
- Sliding vector states that the point of application of the force can be anywhere along the line of action of the force, provided it does not change the effect of the force.
- But when it comes to principle of transmitter of forces, if we are talking with respect to a body, when we apply a force on the body, if the force is in contact with the body, then only there will be some effect.
- Otherwise, there won't be any effect.
- Can be along the line of action of the force, the body would still experience the same effect, provided the point of application of the force is in contact with the body.
- This is the difference between a.
- Sliding vector and principle of transmitter of forces.
- Next, this is a very important slide.
- There are several problems that we will be solving with respect to what you mean by projection and what you mean by components.
- I have explained both of this earlier also, but not as projection.
- Projection is the cosine component of the force on any axis or any plane.
- We will draw a line from the extreme end of the force.
- to the axis B such that this line that is R from the end to this B axis, whatever line that is drawn, that line will make it 90 degrees with the B axis.

Example or application from the lecture

- For example, if we take this.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000038 · EV0000039 · EV0000040 · +1 source units in pack.json

Core explanation · 10:00–14:59

Till this axis A, such that this line, this line is at 90 degrees with respect to the axis A.

Key details

- The projection is represented as FA.
- And F B.
- It is just the cost component of the force.
- Axis whatever we are considering.
- When it comes to components, components are splitting up of the force along the axis such that.
- This would this component when it is acted on a body would produce the same effect as that of the single force.

- We are projecting the force on some axis.
- That's all.
- Just like how we watch a movie in a theater.
- It is projection, but in a movie there will be only one light source.
- Which will fall on the film and an image is formed on the screen which will be larger than the film itself.
- Whereas in case of projection,.
- There will be n number of light sources, but these light sources will always be at 90 degrees to the screen.
- Always it will act at 90 degrees to the screen, and we will take the light source which will pass through the edges of the force and whatever the image that is formed.
- Will be either equal to the length of the force or to be lesser than the length of the force, irrespective of the position of the screen or the axis on which we are considering.
- I hope it is very clear what is the difference between.
- projection and component.
- Very simple, use either law of triangle of forces or law of parallelogram of forces.
- Component of the force R on A axis and B axis by considering it as a parallelogram.
- FA and F B are the projection, whereas F1 and F2 are the components.
- The sides that we are going to get or the components.
- When you use law parallel of forces or of triangle of forces, you get the component of the force.
- But when you simply draw a line which is perpendicular to the axis, you get the.
- Projection of the force onto that axis.
- I hope this is very clear.
- We'll be solving a lot of problems on this, so you need to understand this very carefully.
- We had already discussed about resultant, this slide indicates how do we get the resultant of two forces which are parallel to each other.
- When we have a force F1 and F2, we want to find out the resultant of these two forces.
- In the positive direction and in the negative direction also.
- Find out the result end of these two.
- When we clog these two, this F1 and minus F1 will get cancel.
- If I find out the result end of this, I would get the result end as R.

Example or application from the lecture

- Now here in the first diagram they have given you how to get the.
- Consider F1 and F, find out the resultant of these two as R1.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000042 · EV0000043 · EV0000044 · +1 source units in pack.json

Core explanation · 15:00–19:59

This is one way of solving.

Key details

- Of course, you can go ahead and use vector addition as F1 vector plus F2 vector, the dot product, okay, you will get the resultant of this two forces.
- I had discussed about components where there was axis A and access B.
- I had also discussed this earlier when I talked about resultant how to determine the resultant or use the dot products.
- How do we use the dot product to get the resultant?
- When I explained that I have already considered.
- The x axis and y axis I have discussed about ij what are these unit vectors along x and y.
- How do we get the components f x and f y?
- Now the components and the projection when the two axes on which we have to find out the component and the projection are mutually perpendicular to each other or mutually perpendicular to each other, there is no difference between projection and component because.
- When I try to draw a parallel, this forms a rectangle.
- A rectangle is also a type of parallelogram.
- Rectangle is a parallel, but parallel is not a rectangle.
- Please remember that.
- If I use the trigonometry here, I will get Fx, or if I use parallel law, I will also get Fx.
- Either projection or law of parallelogram of forces or law of triangle of forces to get the components.
- The magnitude, whatever we get, would be the same.
- However, to get the sense, we have the force.
- F1 acting like this.
- Whatever sense that we get, we I rotate this and merge it here.
- I get the sense to be this.
- That would be the sense of your Fx.
- Then I will have this force, I rotate this and merge it with this, I get the sense of FY.
- You can rotate it by between 0 and 90 degrees, not greater than 90 degrees.
- That you have to remember.
- The point of application will be the same as that of the force F itself.
- Next, these are several.
- indication of the diagram which indicates X and Y taken other than the horizontal and the vertical axis.
- The angle made by the force with the x axis, when I draw an x axis here, this total angle will be beta.
- And here also it is if I draw a line x-axis here.
- And extend that this angle made by the force with x axis will be total angle is one spite that is 180 minus beta I get this angle.
- Cause of pi minus beta, I'll get y, and it is it is negative f y.
- And f x, if I rotate this and merge it here, I get x to be this.
- The other things you can do it, I think.
- If there are four two forces that is acting, how do we get the resultant is what is indicated.
- F1 as F1x.
- And F2 as F to X, add these two kit to get the component of the resultant or the force R.

Example or application from the lecture

- Let us look at this rectangular compound.
- Everywhere they have given beta.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000046 · EV0000047 · EV0000048 · +1 source units in pack.json

Core explanation · 20:00–24:59

These are the several steps that you have to follow simply.

Key details

- You can draw n number of diagrams till you understand.
- Then state the governing principle.
- What are the principles that you are going to use?
- Make the calculations, whatever calculation that you're doing.
- Ensure that your calculations are consistent with the accuracy justified by the data.
- Be sure that you have used consistent units.
- Ensure that your answers are reasonable in terms of magnitude, direction, common sense, etcetera Use that to get and then draw the conclusions.
- This is what we follow when we have to solve problems.
- We'll continue in the next class.
- My name is Dr.
- Sventesh from PS University.
- See you in the next class.

Example or application from the lecture

- State the given data, state the desired result that you want, state any assumption and approximation that you are going to make while solving the problem.
- If the data is given as two decimal points, three decimal points, you have to also get the result in two decimal points or three decimal points.
- If the units are given in kilonewtons for one force and newtons in the other force, make sure that you will convert the kilonewt into Newton or Newton into KN and use appropriately.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000050 · EV0000051 · EV0000052 · +1 source units in pack.json

5 UE26ME142A_Unit 1_Class 2.1_AVSummary

Study focus: UE26ME142A_Unit 1_Class 2.1_AVSummary. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000054 · EV0000055

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

My name is Dr.

Key details

- Venkadesh from Civil Engineering in the Department of Science and Humanities under PS University.
- This is class number three.
- In the last class, we had discussed few of the basics that is required with respect to mathematics for understanding the engine mechanic statics.
- Which are listed here.
- Again, the same slide.
- Then we also discussed about what is the difference between deformation, displacement, and deflection.
- Then we defined what is meant by resultant and what are the different coordinate systems three-dimensional, three D and two-dimensional two D.
- Then we also discussed about what is meant by translatory motion and what is meant by rotatory motion.
- Next, we should understand what is called as law of triangle of forces.
- Taken in the opposite order will give the result of these two forces.
- What does this mean?
- These are the two forces which are acting on a particle.
- And these two forces are represented on the sides of a triangle taken in order.
- What does that mean?
- V1 force sense is moving in this direction and it is V2 force is coming like this.
- On the adjacent sides of the triangle, which are represented, the forces have been represented.
- Which is being represented as V dash, which indicates the resultant of the two forces that is V1 and V2.
- But if you observe this sense of this V dash is in the anticlockwise direction.
- In the clockwise direction, V2 is also moving in the clockwise direction, whereas V dash, which is a resultant of V1 and V2, is in the opposite sense.
- sites or the forces, then that gives you the resultant of the two forces V1 and V2 with respect to the magnitude of the resultant and the direction or the sense of the resultant.
- Next we need to understand what is meant by law of parallelogram of forces.
- Law of parallel of forces states that if two forces are acting on a particle, and if these two forces can be represented on the adjacent sides of a parallelogram.
- Then the diagonal of the paragram passing through the intersection of the two forces gives you the magnitude and the direction of the resultant.
- Say V1 is one force and V2 is another force, and if we can represent these two forces on the sides of a parallelogram, so this is one side of the parallelogram, that is V1.
- Whereas this is another side of the paragraph representing V2, okay, these two are the sides of the paragraph, then the diagonal passing through the intersection of the two forces.

- What is the intersection of these two forces?
- Which is a diagonal, this is the diagonal passing through the intersection of these two forces, passing through the intersection of these two forces would give you the resultant of the two forces.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000054 · EV0000055 · EV0000056 · +1 source units in pack.json

Core explanation · 05:00–09:59

And this is the diagonal of the parallelogram.

Key details

- But there are two diagonals.
- This also would be the diagonal.
- But this diagonal does not give you the resultant.
- Only the diagonal which passes through the intersection of the two forces would give you the.
- Resultant of the two forces.
- Here you can see that the two forces are taken such that both of them are moving away from the point.
- You cannot take as I discussed in the law of triangle of forces, both of them are having or will rotate in the same direction.
- No, you cannot do it in case of law of parallel forces.
- If you want to.
- To it or find out the resultant using law of parallel of forces, then you should take both the forces either moving away from the point or toward the point, both of them.
- Then only it is possible for you to determine the resultant of two forces.
- If you take this force to be acting here.
- Then the sense of the force should be towards that.
- If the sense is towards this point itself, then this would be the diagonal which represents the result.
- Kindly understand this carefully.
- The this problem will not come in law of triangular forces, but in case of law of triangular forces, the two forces that you take should have the same sense.
- It should have or it should rotate the in the same direction.
- Then only you will get the.
- Resultant as the closing side of the triangle in the opposite sense.
- In case of law of parallel ram forces, the two forces should either move from each other or should move towards each other.
- Then you can take the resultant as the diagonal passing through the intersection of these two forces.
- By taking this as the force.
- And this, then what happens?
- Here in the downward direction here, so that both of them will have the same sense, then the closing side will give you the resultant.

- I hope you have understood this.
- Next is in fact I told you that forces are vectors.
- And I also told you if you remember in the very first class, you should know what is the difference between a dot product and a cross product.
- A cross product is nothing but.
- The multiplication of the position vector of the force into the force represented in terms of vectors.
- It means that.
- It is the component of the force along the x axis, say as f x and i, where i is the unit vector of the force along x direction, then f y.
- J where f y is the component of the force along y direction and j is the unit vector of the unit vector along y direction.
- If it is three dimensional, you will get all the three axis, so you get it as FXI, and if I assume that all the three will be positive, then it will be FYJ plus F Z K.
- Where Fx, FY, and FZ are the component of the force along X, Y and Z direction.
- Whereas Ij K are the unit vectors along the axis X, Y, and Z.
- Then what is position vector?
- Position vector is the component of the point from the starting of the force to the end of the force along x, y, and z, which is represented as x i plus yj plus k.

Example or application from the lecture

- If it is a law of triangle of forces, if it is long parallel forces, you have to consider it like this.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000058 · EV0000059 · EV0000060 · +1 source units in pack.json

Core explanation · 10:00–14:59

z where or z k where x y and z are the component of the position of the vector along x y and z axis and your i j k or unit vectors along.

Key details

- X, Y, and Z axis.
- We should remember this.
- I cross I is equal to 0.
- That means you cannot cross multiply the.
- Position and the magnitude of the component which are in the same axis that is x axis.
- If it is Xi, cross FX I, then you get it as I cross I, I cross I is 0.
- K cross k is also zero, j cross j is also zero when it comes to cross product.
- Whereas I cross j will give you k which is positive.
- J cross K will give you positive I and K cross I will give you positive J.
- Whereas if it is I cross K you get as negative J, if it is K cross J, you get us negative I.
- If it is J cross i you can as negative k.
- You have to remember this.
- What is scalars?

- Scalas are such quantities.
- Which have only magnitude, but they don't have direction.
- What are vectors?
- Vectors are physical quantities which will have both magnitude and direction.
- I had also talked about when I talked about characteristics of the forces in the very first class, I told you according to the textbook,.
- What are these characteristics?
- I'll repeat it.
- One is the magnitude of the force, the direction of the force.
- The point of application of the force and the sense of the force.
- The point of application is here where the force is acting.
- In the textbook he combines this with.
- Based on the point of application of the vector, we can divide the vector into three categories, namely free vector.
- Sliding vector, fixed or bound vector.
- What is this?
- What is a free vector?
- A free vector is a vector in which the point of application of the vector can be anywhere in the space.
- But it will not have any difference on the effect of this vector on a body.
- If the point of application of the vector gets changed to any point in space, the effect will not change.
- Such vectors are called as free vector.
- I will talk about it when I take up the vectors.
- When I take up the couple.
- A sliding vector is a vector in which the point of application, that is this point of application, can be anywhere along the line of action of the force.
- Only along this I can change the point of application.
- Either I can take it as this.
- Or this, or this, or this, or this, anywhere I want.
- But I cannot change it anywhere I want.
- Only along this line I can shift.
- Then the effect of this vector on the body will not change.
- But if I change it anywhere else, the effect is going to change.
- Such vectors are called as sliding vector.
- Next, coming to fixed or bound vector, fixed or bound vectors are vectors in which there is only one vector.

Example or application from the lecture

- Alright, next let's look at what are the different types of vectors.
- For example, distance, speed, etc.
- For example, velocity.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000062 · EV0000063 · EV0000064 · +1 source units in pack.json

Core explanation · 15:00–19:59

point of application for a given vector.

Key details

- Along the line or anywhere I want, the effect is going to change.
- Sliding vector, the point of application can be chang only along the line of action of the force, then the effect is going to be the same.
- Whereas fixed vector the point of application is only one, only then the effect is going to be the same.
- If I change the point of application anywhere, the effect is going to change.
- We take x-axis horizontal and y-axis vertical.
- The intersection of these two is taken as origin.
- The point which we have to plot along X, if it is positive, we take it from the origin to the right.
- That is positive.
- How do we plot the graph?
- Similarly, positive y means we will take it above the origin.
- Negative y, below the origin.
- Negative x, left of origin.
- That is along the x-axis.
- This is the usual sign convention for forces.
- But why vice versa is also true.
- If I take the component of the force along X as positive, moving from right to left as positive, then moving from left to right will be negative.
- If the force is moving from left to right along X, it is positive.
- From right to left, it is negative.
- If it is moving upwards, it is positive.
- If it is moving downwards, it is taken as negative.
- What is the difference between a scalar and vector?
- Then we have defined three different types of vectors, that is free vector, sliding vector, and fixed or bound vector.
- Then we have also looked at what is the sign convention for forces.
- What is the usual sign convention that is used for vectors or forces, but you can also change it vice versa is also true.
- This is class number three.
- I thank everybody.
- My name is Dr.
- I'm from the civil engineering under Department of Science and Humanities under PS University.
- See you till the next class.
- The next class is class number four.

Example or application from the lecture

- For example, if this is the point of application of the vector, if I change this point anywhere.
- Next let's look at what is a sign convention.
- Okay, if it is moving to the right, for example, if I take this and the component of this is going to be along this, moving from left to right, we take it as positive, whereas if it is moving from right to left, right to left, we take it as negative.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000066 · EV0000067 · EV0000068 · +1 source units in pack.json

6 UE26ME142A_Unit 1_Class 2.2_AVSummary

Study focus: UE26ME142A_Unit 1_Class 2.2_AVSummary. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000070 · EV0000071

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

Dear student, I welcome all of you for engineer mechanics static classes.

Key details

- My name is Dr.
- Sv Venkadesh from Civil Engineering under Department of Science and Humanities, PS University.
- This is class number four.
- In the last class, we had discussed law of triangle of forces, law of paragraph of forces, cross product, then scalars and vectors, then what are the different.
- Types of vectors based on the point of application of the vector.
- And then what is a sign convention that we use to represent the vectors?
- This was discussed in the last class, that is class number three.
- Cause of the angle made by the two vectors that is V1 and V2, that is this angle.
- plus 2 V1 into V2 into cos of the angle between the two vectors V1 and V2 which I am considering this angle as theta.
- Direction of the resultant that is the angle made by the resultant with respect to one of the vectors, say V1, if this is the angle that I am considering, say alpha.
- Then tan alpha will be equal to we will to get that alpha we will make a construction here.
- This becomes a rectangle triangle, whereas this side will be equal to V1 sorry, it is going to be equal to V two sine of angle theta that is this angle which is same as the angle between V one and V two.
- cos of theta okay.

- The direction of the resultant, the sense of the resultant, and of course the point of the resultant will be the same as the point of application of these two forces, that is this point.
- Or we will go for what is called as rectangular components of a force.
- You this diagram you can see.
- What does this diagram indicate?
- We have a force V which is making an angle of theta with respect to the x axis.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000070 · EV0000071 · EV0000072 · +1 source units in pack.json

Core explanation · 05:00–09:59

No, to get that we will draw a line parallel to Vx, which is.

Key details

- Along x axis passing through the edge of V, that is from here, so an horizontal line.
- Rectangle, and that's why it is called as a rectangular component.
- However, we will get the component of the force V that is Vx and Vy, which is along the x-axis and along y-axis as V.
- Cos of this angle theta, we get Vx.
- These two are mutually perpendicular to each other, and if we have a force V, we can still get the component V x dash and Vy dash as the cause of this angle.
- And V sign of this angle as V y 1.
- We have Vx and Vy and then if we want to get V which is nothing but the result n we go for what is called as a dot product which I will be explaining it later.
- This is the vector representation of a force.
- What is Vx?
- It's the component of the force along x axis.
- Vy is the component of the force along y axis, whereas I and J are the unit vectors along.
- X axis and along Y axis.
- This is for a force which is two dimensional.
- If we have a three dimensional one, then we will also get plus.
- Vx as I have already informed you that Vx is nothing but V cos theta, that is the component of V along X, and Vy is the component of V along y axis, that is V sine theta.
- But if the two axes are mutually perpendicular to each other, we can get the result and by simply doing this one.
- Taken a force V1 and F force V2.
- Whatever that you are going to use, simply V will be equal to V1 plus V2.
- It is a dot product of.
- The two vectors will give us the magnitude of the resultant of these two forces.
- We have a force V1 of 4 units that is the magnitude of the force which makes an angle of 45 degrees with the y-axis.
- The sense of V1 is moving away from the origin, the origin is this, the intersection of the x-axis and the y axis is the origin, the force is moving away from that.

Example or application from the lecture

- Next let us see what this vector addition is.
- Now there is a simple example that I have taken here.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000074 · EV0000075 · EV0000076 · +1 source units in pack.json

Core explanation · 10:00–14:59

We also have one more force which is also moving away from the origin, which is V2.

Key details

- The magnitude of V2 is three units, but it makes an angle of thirty degrees with the x-axis below the x-axis.
- The construction is we have constructed a parallel where there is a line.
- Which is drawn parallel to V1 passing through the edge of V2.
- And according to the law of parallelogram of forces, the diagonal passing through the intersection of the two forces would give us the magnitude both in direction.
- And in magnitude.
- V1 into V2 into cos of the angle between the two forces that is this, which is simply equal to 45 degrees, that is this one, 45 plus 30 degrees, which would be 75.
- That gives us the magnitude of the resultant, yes.
- Or if I take only this one, then this total angle will be equal to 30 plus alpha, 10 of 30 plus alpha will be equal to V1, that is this V1,.
- Sine of this angle, which is the total angle, that is 45 plus 35, which is 75, divided by V2, which is 3, plus this component, which is V1.
- cos of 75.
- That is one way of solving or getting the.
- Magnitude, direction, the sense of course will be when you merge these two, you will get the sense.
- That is one way of determining the resultant of these two forces.
- The other way of determining the resultant of these two forces is.
- Simply determine the component of V1 along X and Y and determine the component of V2 along X and Y.
- The dot product of these two would give us the magnitude of the resultant.
- V1 in terms of the unit vector i and j will be v1 is equal to 4.
- Directly this angle is thirty, so we can see that this is thirty.
- This is 4 into cos 45 i is the 4 into cos 45 is the component of V1 along X.
- I is the unit vector along X.
- Now that would be positive because the sense of this when I rotate it and merge it along x axis, the sense is going to be along X, so that is positive.
- When I rotate V2 and merge it with the Y the X axis.

Example or application from the lecture

- This angle is going to be 45 because they have given this angle as 45.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000078 · EV0000079 · EV0000080 · +1 source units in pack.json

Core explanation · 15:00–19:59

Then I'll get the sense to be same as that of x, so it is positive, so it is 3 into cos thirty degrees, my component of v2 along x, whereas i is the unit vector of along x axis.

Key details

- Next to find out along y, when I rotate this and merge it along y, I get it to be acting downwards, so that's why it is negative.
- Next to determine the component of V2 along y, it is V2 into sine of 30 degrees, so I get it to be acting downwards.
- This would be the component of V2 along y and j is the unit vector along the y axis.
- Of the result, and we simply add V1 and V2.
- The dot product of V1 and V2, we will get the magnitude yes.
- Plus four sine forty five minus three sine thirty degrees J would be the magnitude of the resultant of these two forces.
- I hope you can.
- Find out the magnitude by law of parallel of forces as I have explained, and by resolving the two vectors along their components X and Y, and then adding them or finding out the dot product of these two, and then getting the magnitude of the resultant of these two forces.
- However I will be explaining how to get the.
- Resultant of these two forces.
- The point of application of the force is the origin, that is the intersection of X and Y.
- The force is moving away from the origin.
- By 4, this triangle indicates the slope of this vector.
- We can easily determine what would be this angle, or we know that this is a right angle triangle.
- The hypotenuse is gonna be five, three, four, five method.
- You can use that or you can simply determine this angle by taking this right angle triangle, you can find out this angle theta 1 if I take this angle as theta 1.
- That would be equal to tan inverse of 3 by 4.
- You can find out the magnitude of that.
- Next we have V2 force.
- Of 12 units, the origin is still the intersection of X and Y, and again the force is moving away from the origin.
- Cos theta 1, then this is theta 1, I plus 10 sin theta 1 j.
- Next coming to V2, the component of V2 along X would be in the negative X direction.
- Whereas for the y, if I rotate this and merge it, I get the sense of the component of this force and the positive y direction, so it will be plus, so that will be plus 12 cos 60 degrees J.
- Take the dot product of these two, so you will get the magnitude of the result end.
- We'll stop at this point and we will continue in the next class.
- And this is the fourth class.
- I'll come back to this problem formulation and obtaining solution in my next class.

- However, explain how to get the this.
- So we will stop here and we'll continue in the next class.
- And my name is Dr.

Example or application from the lecture

- The inclination of V one is given as three.
- The inclination of this V2 with the negative x axis is given us sixty degrees.
- One is by using law paragraph of forces, the other one by resolving the given forces along their components and then getting the results.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000082 · EV0000083 · EV0000084 · +1 source units in pack.json

7 UE26ME142A_Unit 1_Class 3_AVSummary

Study focus: UE26ME142A_Unit 1_Class 3_AVSummary. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000086 · EV0000087

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

My name is Dr.

Key details

- Svink Tish, Civil Engineering, Department of Science and Humanities, PS University.
- This is class number six.
- In the last class, we had discussed about the a difference between components and projection.
- How do we determine the components?
- How do we determine the projection?
- What is rectangular?
- And several definitions that we were supposed to know to solve problems of engineer mechanics.
- In this class we will look at what are the problems that we can solve.
- We will try to solve one unsolved problem also.
- The statement of the problem is the forces F1, F2, and F3, all of which act on point A of the bracket.
- Are specified in three different ways.
- Determine the X and Y scalar components of each of these three forces.
- And what is asked is to determine the X and Y scalar components, only the components of the forces along X and Y.
- This is point A.

- He says three different ways.
- The direction of the forces is specified in three different ways.
- What are these three different ways?
- One is for.
- They have mentioned the angle made by the force with respect to the x axis.
- X axis is horizontal, y axis is vertical.
- This we have to make out from the diagram.
- Or this says a line which is parallel to X axis.
- The vertical distance is three, the horizontal distance is four.
- Or we can also determine what is the angle made by F2 with respect to x axis or y axis.
- Say with respect to x axis, it will be equal to tan inverse of 3 by 4.
- The direction of the force F3 is not mentioned.
- Instead, they have mentioned the starting point of F3, the ending point of F3 as B, A and B, starting point is A, ending point is B.
- from the origin as point two along x axis.
- So first force F1 is indicated here.
- This is F1 which makes an angle of thirty five degrees with the x axis.
- It should be equal to F cos 35 degrees straight away and F1y will be equal to F1 sine 35 degrees.

Example or application from the lecture

- I've taken a few example problems to understand how do we go about solving these problems and let's see depending upon the time.
- We should read the diagram that is given.
- From the statement of the problem, what is given is there are three forces which is acting at point A, that we can make out from the diagram.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000086 · EV0000087 · EV0000088 · +1 source units in pack.json

Core explanation · 05:00–09:59

To determine F2x, if I take this.

Key details

- It is gonna be equal to F two cause of this angle, cause of this angle is 4 divided by 5 into 500 will give me F2x.
- F3 is acting here.
- Now if I take that angle made by the force F3 with respect to y axis as alpha, I can easily calculate what is the angle alpha as tan inverse of 0.2 by point four because this forms a right angle triangle.
- Adjustant opposite is 0.2, adjustant is 0.4, so I can get alpha.
- These are the scalar components of the forces F1, F2, and F3.
- which act on the fixed structure at B into a single equivalent force R.
- Resultant of the two forces that is force P and T, what is the resultant of these two forces?

- We can solve this in two different approaches, or whichever approach you want, you can take several approaches.
- Parallel of forces, the other one is using the law of triangle of forces, and one more is by resolving P under T.
- Along X and Y, and then combining out finding out the dot product of this, you will get the result.
- You can see that P is represented here, which is 800 newtons.
- forms or this two forces are represented now on the adjacent sides of a parallelogram.
- Both of them are moving away from point B.
- And how do I find this magnitude of the resultant?
- That would be equal to P square.
- Plus T square plus 2 times P into T into cos of the angle made by these two.
- How do I find the angle?
- or rather 180 minus alpha.
- I can find out what is the result and r from that.
- The two forces are taken on the adjacent sides of a triangle, taken in order, you can see that this is in order.
- The closing side of the triangle taken in the opposite order gives us the resultant.

Example or application from the lecture

- Now let us move on and take one more example.
- This example the statement of the problem states that combine the two forces P and T.
- Let's look at all these three different ways of solving the problem, one at each time.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000090 · EV0000091 · EV0000092 · +1 source units in pack.json

Core explanation · 10:00–14:59

The other way of solving this is by considering or finding out the component of the two forces along X and Y, and then finding out the resultant R.

Key details

- How do I get Rx square plus Ry square?
- I get the magnitude of R.
- R X coming this, R Y coming this.
- When I merge these two, I get the resultant force acting like this.
- This is how we can determine the resultant of the of these two forces.
- Vector approach to solve.
- In the two previous problems that I have explained, what we have looked at is how do we solve it by going ahead and solving it in scalar approach.
- The 500 newton force F is applied to the vertical pole as shown.
- This is a vertical pole.
- A 500 newton force that is this one is applied to this vertical pole as shown.
- This is acting at point A, which makes an angle of 30 degrees with the.

- Y-axis or with the pole, the forces are acting at an angle of thirty degrees.
- The magnitude of the force is 500 newtons.
- Write F in terms of the unit vectors I and J and identify both its vector and the scalar components means what?
- We have to find out the component of the force f along x and along y.
- How do I make out that?
- It is i and j.
- Then we have to determine both scalar and vector approach.
- Scalar means only Fx and Fy, we have to determine that.
- The second part of the problem is determine the scalar component of the force vector along the x-dash and the y-dash axis.
- Which is the x-dash axis?
- The axis along the force itself is x dash and perpendicular to that is y dash.
- You have to find out the component of the force along this x dash and y dash.
- That is second one.
- The third one is determine the scalar components of force.
- Yes, along the x and the x-dash axis, that means what?
- I have to find out the scalar component of this force along x and along y-dash axis, along this axis.
- That is what we have to find out.
- So first thing we have taken the first case where we have to determine the.
- A component of the force along X and Y.
- The component of the force is also represented along X and Y.
- Whereas F sine sixty degrees with a negative sign, I'll get it as F y.
- Why it is negative?
- Because the sense of the force F y is acting downwards, so I get it as negative.
- Now to express this as a vector, it is nothing but fx i minus f y j because f y is acting downwards, so it is negative.
- The scalar component is just fx which is equal to f cos 60 degrees.
- The scalar component of y that is f y is equal to f cos 60 degrees, that's it.
- The next one is to determine the component of the force along x dash and y dash axis.
- The component of the force along Y will be F, cos ninety degrees, and we know that cos ninety degrees is.

Example or application from the lecture

- Let's go at and try to solve another problem.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000094 · EV0000095 · EV0000096 · +1 source units in pack.json

Core explanation · 15:00–19:59

Next coming to determination of the scalar component of the force f along x and y dash axis.

Key details

- Where you rotate it and merge it up to ninety less we can rotate it.
- Parallel of forces.
- They have asked you to determine the components.
- Component is when we have to determine by using law of triangle of forces.
- Component will produce the same effect as that of the force.
- We can determine the component of F along F x and FY.
- For that, we have to determine the angles.
- This is parallel to this, so this forms the same angle, 60 degrees and 60 degrees.
- Ah whereas these two will be the same, so they are alternate angles, so 90 and 90.
- We know that this forms a triangle.
- by sine 90 is equal to f pi sine 30.
- By sine 60 will be equal to f by sine 30, so I can get f y dash.
- The statement of the problem is forces F1 and F2 act on the bracket as shown.
- Determine the projection of FP of their resultant R onto the B axis.
- Clearly understand the problem.
- What is asked is there are two.
- This is axis A, this is axis B.
- They have asked you to determine the projection of these the resultant of these two forces.
- First we have to find out what is the resultant of these two, then projection of the resultant onto the B axis.
- How do I construct it?
- We have this force and this force.
- We know this angles.
- The total angle is 50 degrees.
- That is 20 plus 30 is 50.
- Is along A and this force F2 is along B.
- This is parallel to this.
- this edge, so these two lines when I draw, it was going to cross each other at this point.
- The projection of R onto the B axis.
- Touches B axis such that this angle is at 90 degrees.
- That would be equal to F P plus this value.
- This value will be equal to F1, which is same as this.
- Cause of this will be this value.

Example or application from the lecture

- Forces F1 of magnitude 100, F2 of magnitude 80 newtons, the sense of the two is given, the direction of the forces are also given as 30 degrees with respect to X, 20 degrees also with respect to X.
- Let's look at this.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000098 · EV0000099 · EV0000100 · +1 source units in pack.json

Core explanation · 20:00–24:59

Which is equal to 144.3 newtons.

Key details

- That would be the projection of the resultant on the B axis.
- I have explained it here.
- F2 square plus F1 square plus 2 times F1 F2, cause of the angle between the two.
- This angle which is 50 degrees, I'm sorry I have made a mistake here.
- I have taken it as 130, it should be 50 degrees.
- I hope this is very clear.
- We will come back to this because I'm shooting the time.
- We will look at this problem in the next class.
- my name is Doctor Svang Tesh.
- We'll meet in the next class, class number seven.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000102 · EV0000103 · EV0000104 · +1 source units in pack.json

8 UE26ME142A_Unit 1_Class 4_AVSummary

Study focus: UE26ME142A_Unit 1_Class 4_AVSummary. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000106 · EV0000107

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

I am Dr.

Key details

- SV Venkitesh, Civil Engineering, Department of Science and Humanities, PS University.
- In the last class, that is class number six, we had discussed several problems and how do we solve them by both scalar approach as well as vector approach and they were solved problems.
- The statement of the problem is like this.
- Two forces are applied to the construction bracket as shown.
- Determine the angle cheetah, which makes the resultant of the two forces vertical.
- Determine the magnitude R of the resultant.
- And they have also asked you to determine the magnitude of the resultant R.
- No down, this is the angle made by the force F1.
- With respect to X, okay.
- The force F1 is in the plane of X and Y only.
- In the XY plane itself.

- Now to get the result of these two forces, first thing that I'm going to do is I'll resolve both the forces F1 as well as F2 along its components, that is along x-axis and y axis.
- F1x and F1Y.
- I get F1x as F1.
- cos 70 degrees f1 y as f1 sinilarly for this f2 x will be equal to f2 cos theta and f2 y will be f2 sine theta no adding up of.
- The components along x axis, okay, that is the component of f1 along x, that is f1x, and the component of f2 along x, which is f2x, if I add these two, that should be.
- equal to zero.
- Why should that be equal to zero?
- Because what are we supposed to find out is the angle theta which makes the resultant to be vertical.
- It should be 0.
- Because F1 is acting along X, it's positive.

Example or application from the lecture

- From the diagram, what is that we need to look at is they have given you the magnitude of the force F1, which is 800 newtons, and they have also given you what is the direction of this force F1 with respect to the negative x axis, which makes an angle of 70 degrees.
- Next, they have also given you the magnitude of the force F2, which is equal to 425 newtons, and the angle theta which we have to determine is the angle made by the force F2 with respect to x axis again.
- Even though you see that the diagram that is given, the bracket that is given is three-dimensional, but the forces acting is in the same plane that is XY plane.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000106 · EV0000107 · EV0000108 · +1 source units in pack.json

Core explanation · 05:00–09:59

The component of F1 acting along X is positive.

Key details

- The component of the force.
- Along X of F2 is negative because the sense of it is opposite to X.
- By solving it, we are going to get the value of theta which is 49.92.
- Then, so that is what we have to find out in the first case, theta.
- Theta is what we have got.
- The second case, they have also asked you to determine the magnitude of R.
- Find out the summation of that.
- That will be equal to the resultant force.
- And the force, the resultant force will be acting along y only, so you get that value.
- Both F1 Y and F2 Y both of them are acting downwards, so it will be negative.
- The component of F2 along y will be minus F2 into sin theta, which is equal to minus r y, which is equal to minus r.

- That is the resultant.
- I hope all of you have understood this problem.
- Now we'll move on to the next problem.
- The next problem that I've chosen is 2 bar 17.
- The statement of the problem goes like this.
- On the TB.
- Read out from the beginning.
- The two forces shown at in the XY plane of the T beam cross section.
- A line of action that lies 15 degrees above the negative x axis.
- What is this?
- The resultant will be acting somewhere here.
- Resultant, of course, the point of application will be the same as the point of application of the two forces.
- Or F1, which is which we have to find out, or rather, what is the direction of the force F1, which makes an angle of 30 degrees with the y-axis, then what is the magnitude of F2, which is 2.8 kilonewtons.
- And the angle made by F2 with respect to x axis is theta.
- We have to determine the magnitude of F1 and we have to determine the direction of F2.
- That is with respect to x axis.
- F2 is acting towards the origin, the magnitude of that is 2.8, and it makes an angle of theta with respect to the positive x-axis.
- Excuse me, F1 is also acting towards the origin.
- It makes an angle of 30 degrees with the negative y axis.

Example or application from the lecture

- Obviously, this is acting downwards, only the magnitude is given here, and of course it is acting downwards.
- The two forces shown act in the XY plane, they have already given that it is acting in the XY plane.
- Even the diagram that is given is three-dimensional, but since they are told that the forces acting is in the XY plane, we can treat this as a two-dimensional force system.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000110 · EV0000111 · EV0000112 · +1 source units in pack.json

Core explanation · 10:00–14:59

Magnitude is 3.5 kilonton.

Key details

- Alpha is the angle made by the resultant with respect to the negative x-axis of the state is above, that is 15 degrees.
- What are we supposed to find?
- Is the magnitude of F1 and the angle theta of the force F2 with respect to x-axis.
- F2 also moving away and the result end also.

- How is it 45?
- 90 minus 30 minus 15.
- will be forty five degrees.
- Using the law of triangle of forces, so F1, F2 taken in order.
- The magnitude of the force F2 as 2.8, and we are supposed to find out the magnitude of F1.
- We know the angle made by F1 with the resultant R that is 45 degrees.
- We can also determine.
- What is the angle made by the force F2 and the resultant which will be equal to theta plus 15?
- How is it?
- It's theta plus 15 here.
- When I put it here the this would be the alternate angle for this which will also be equal to theta plus fifteen.
- It states that 3.5 divided by sine of the sine of this angle, that is sine of 120 minus theta, should be equal to.
- Magnitude of this force which is two point eight, opposite angle as forty five, so sine of forty five should also be equal to this force F one.
- Angle opposite to that is this which is.
- Equal to theta plus 45 sine of theta plus 45.
- And 3.5 sine 120 minus theta.
- If I take these two equations, I can rewrite this equation as sin of 120 minus theta equals 3.5 sine 45 divided by 2.8, from which I can find out the value of theta.
- By simplifying it.
- Once I get the value of theta, I can take these two equations, okay, substitute the value of theta here, and then find out the value of f1.
- I have not gone for resolution of forces F1 and F2, because when you go for resolution of forces, the solution will be a lengthier one.
- The law of triangle of forces or using the sine rule.
- Law of triangle of forces is nothing but which we have got it from sine rule.
- I've taken one more problem, 2 bar 22.
- The statement of the problem goes like this.
- A sheet of an experimental composite is subjected to a simple tension test.
- To determine its strength along a particular direction, the composite is reinforced by the Keplar fibers shore.
- And a close up showing the direction of the applied tension force F in relation to the fiber directions at point A is shown.
- If the magnitude of F is 2.5 kilonewt, determine the components FA.

Example or application from the lecture

- Let's look at the solution.
- Okay, let's go on to the next problem.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000114 · EV0000115 · EV0000116 · +1 source units in pack.json

Core explanation · 15:00–19:59

And F B of the force F along the oblique axis, A and B also determine the projection P A and P B of the force F onto the A and B axis.

Key details

- If you take a small enlarged view of the intersection of the fibers, it is got to look like this.
- What are we supposed to find?
- The component of the force F along axis A and B, as well as the projection of the force along A and B.
- That is, this is the axis B and the force F is acting along X.
- I hope it is clear.
- we have got the angles, so now using the sign rule to get the components.
- Components I have told you will follow law of triangle of forces or law of parallelogram of forces.
- The law of parallel of forces or the law of triangle of forces is derived from sine rule, so we can use sine rule to get it.
- I can write it as FA, which is the force along A divided by the sine of the angle between the other two.
- Divided by the sine of the angle between the other two, which is x-axis and a that is sine of 40, which should be equal to f, which is the fourth sine of the angle between the other two, which is the total angle between A and B, which is sine 50.
- I can get the value of FA and FP.
- The units of the forces are important.
- Kilonewtons, so you will have the components of the forces also in kilonewtons.
- Next coming to the projection.
- Projection is the cause component of the force.
- Here you can see the projection, projection of PA, that is projection of the force F.
- On A, how do I get?
- If this is A, I should draw a line from the edge of the force to the axis A, such that this line would make an angle of 90 degrees.
- F cos 40 would give me the projection of the force on A.

Example or application from the lecture

- What is given is you have a diagram indicating the composite sheet to which a force F is applied, a tensile force F is applied.
- They have also given you what is the angles made by these fibers with respect to the axis.
- The tensile force is given as 2.5.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000118 · EV0000119 · EV0000120 · +1 source units in pack.json

Core explanation · 20:00–24:59

That is what is written here.

Key details

- F cost 40.
- This angle is 90 degrees, so this forms a.
- Okay, so we will stop here because we are running short of time.
- We'll continue in the next class.
- My name is Dr.
- SV Venpej from Civil Engineering Department of Science and Humanities PS University.
- We'll meet in the next class, that is class number eight.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000122 · EV0000123 · EV0000124 · +1 source units in pack.json

9 UE26ME142A_Unit 1_Class 5_AVSummary

Study focus: UE26ME142A_Unit 1_Class 5_AVSummary. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000126 · EV0000127

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

I am Dr.

Key details

- Venkitish from Civil Engineering under Department of Science and Humanities, PS University.
- This is class number eight.
- Now today we'll continue with that and we will discuss the next topic, which is on moment.
- What is a moment?
- The definition of a moment is nothing but.
- The rotating effect produced by a force on a body.
- When a force is applied on a body, if the body rotates, that rotation of the body is called as a moment.
- One is translatory motion, the other one is the rotatory motion.
- The translatory motion is the displacement.
- When a force is applied onto a body, the body would move from a point A to a point B, or when a force is applied, the body would rotate.
- Without any displacement, it will be at the same point, but it will rotate.
- The rotating effect caused by the force is termed as moment.
- Mathematically, we represent moment as M is equal to R into F.
- or F into R if we are solving it by scalar.
- But if we are solving it in vector approach, then it follows what is called as a cross product.

- In earlier classes I had also discussed about cross product.
- I have talked about why it should be the position vector cross force vector and not force vector cross position vector.
- The distributed property is not valid in case of cross product.
- Position vector cross the force vector.
- Now in scalar approach it can be F into R or R into F, it does not matter because there we have to take the rotation into account and then decide whether it is positive or negative.
- But when we are using cross product.
- By doing or executing the cross product, we would get the sign for the momentum.
- Ringe is an instrument which is used to rotate a pipe or a bolt or a net.
- Which is perpendicular to this tool, which is called as a range.
- The magnitude of the rotation is going to be equal to the force that is applied into the perpendicular distance between the point where we want the moment.
- And the point where the force is applied.
- That distance in this case is B.
- Has to be 90 degrees, then only this distance would be the perpendicular distance.
- If you orient your thumb, right hand thumb.

Example or application from the lecture

- In the last class we had solved a few problems on resolution of forces, splitting up of a given force along its ax along its x and y axis.
- And determining the resultant of the given force.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000126 · EV0000127 · EV0000128 · +1 source units in pack.json

Core explanation · 05:00–09:59

Along the positive axis on which or about which the movement is caused, and if the other four fingers, as you can see, is folded like this.

Key details

- If that rotates, if that rotates, then.
- Like this, okay, as indicated in this diagram, then that would be the positive sign for the moment also.
- Orient the thumb, orient the thumb along the positive direction of the axis about which the moment or the rotation is occurring.
- If the rotation occurs similar to the other.
- Fingers of the right hand as indicated here, then we take that as positive.
- That is this axis.
- I want to find out what is the moment.
- If this distance is R, this angle is alpha, either I can resolve this force.
- Which is perpendicular to the axis and parallel to the axis, only the component of the force which is perpendicular to this axis would produce a moment.
- F sin alpha into R would be the magnitude of M, that is one way of solving it.

- The other way of solving it is you extend this line and determine what would be the perpendicular distance of this force.
- From point A, which is indicated here as D.
- This is 90 degrees.
- If this is alpha, then this would also be equal to alpha.
- We know that this is R.
- This forms a right angle triangle.
- Whether you resolve the force or you determine the perpendicular distance of the force with the point about which it is rotating.
- Or about which we are finding out the moment, we get the same value.
- Force is F and the perpendicular distance is D.
- The moment produced is the force into this perpendicular distance D.
- If this is x-axis and this is y-axis, the body is in xy plane, then the movement would be produced in an axis which is perpendicular to xy.
- The thumb of my right hand towards me, then you can see that the other four fingers are rotation.
- That rotation will be the positive direction of the moment, and that is how it is rotating.
- Waring Nord's theorem states that the moment produced by a force is equal to the moment produced by the force.
- Components about the same point.
- I'll repeat.
- Moment produced by a force, in this case, if I take R as the force.
- The movement produced by this R about any point, now I will take point O.

Example or application from the lecture

- If the rotation occurs opposite to this, that is like this, then we consider it to be negative.
- Say I will take the same example, an axis is there.
- Next, let's look at what is Warring Nord's theorem.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000130 · EV0000131 · EV0000132 · +1 source units in pack.json

Core explanation · 10:00–14:59

That should be equal to the component of this force R, which in this case is taken as P and Q about some axis.

Key details

- We cannot represent a force by about any other axis which will produce the same effect.
- No, at least there should be two axis about which we can find out the components.
- The moment produced by this R about O should be equal to the moment produced by Q about O plus the movement produced by P about O.
- It is the algebraic sum.
- Not just the sum, it is algebraic sum.
- B you have a movement produced by R above O will be R into B.

- That would be equal to your now.
- How what is the sense of this?
- The R is acting like this.
- The point O is here, so it makes a rotation.
- Like this, that is clockwise rotation.
- Clockwise rotation we take it as negative, and anti-clockwise rotation we take it as positive.
- Or vice versa, you can take it vice versa also.
- But usually when we go for vector approach, that is the cross product, we get clockwise as negative and anti-clockwise as positive.
- That should be equal to your P into the perpendicular distance of P.
- This is the perpendicular distance, which is considered as small Q.
- This force P and this perpendicular distance are small P would produce an anti-clockwise movement.
- That is what Varingov's theorem states.
- What is Varinghaus theorem states?
- Moment produced by a force about any point is equal to the moment produced by the force components about the same point.
- The statement of the problem is calculate the magnitude of the moment about the base O of the 600 newton force in five different ways.
- How do I get the moment of this force, 600 newtons, about this point O in five different ways.
- Then if you read the diagram.
- This diagram gives you the position of A.
- It is at a distance of 2 meters.
- If I take this as x-axis.
- And this has y-axis, okay.
- 2 meters positive x and the origin O is 4 meters from the intersection of these two lines.
- From O to A, the horizontal distance is 2 meters.
- Then the force is inclined at an angle of 40 degrees and the sense of the force is moving away from A, like this.
- This is the same line of action of the force.
- We have to take a moment about this.
- One way is take this force, determine what would be the perpendicular distance of this force from O.
- If I can find that then that will be equal to 600 into D, that is one way of solving it.
- The next way of solving it is resolve this force along this and along this.
- Take movement of this about O, take moment of this about O.
- Next, extend this line and see to it that it cuts this axis at some point.
- Resolve this force along horizontal.
- Determine what is the distance from here to here.

Example or application from the lecture

- We can resolve a given force along at least along two axis, not lesser than that.

- To have the consistency, let us follow the same thing, we will have anti-clockwise as positive and clockwise as negative.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000134 · EV0000135 · EV0000136 · +1 source units in pack.json

Core explanation · 15:00–19:59

Then you can take this for the horizontal component of the force and distance from here to here.

Key details

- You can also extend this line downwards till it touches the horizontal line at O.
- Then resolve this force along vertical.
- Determine what is the distance from here to here.
- Likewise, you can solve or find out the moment of this force.
- About the point O.
- See first method that I had talked about is this.
- You have 600 newtons, find out this D.
- That's one way of solving it.
- The other way of solving it is indicated here.
- You can resolve the force along X and Y as F1 and F2.
- You already know the distance 2 meters and 4 meters using this the 600 sine 40 into 2.
- Would be clockwise negative.
- This 600 cost 40 will also be clockwise into those four.
- Combination of these two would give you the moment about O.
- That's one way another way of solving it.
- Now next one is you extend this till it touches this axis.
- Then you resolve the force as we have done it here into F1 and F2.
- You will get F1 into this distance, if I can find this distance, okay, which is indicated as D1, then you can get the moment which will be clockwise.
- Next one is you extend this till it touches this horizontal line.
- Find out what is D2, then your F2 into D2, you will get the moment about O.
- This is one more way of solving it.
- This is another way of solving it.
- You can also try and solve it in one more way.
- Whatever you feel it is correct.
- I am not going to talk about it right now, but if you have any doubts, you can always meet me and then I will explain about that.
- The problem number is 2 bar 32.
- The problem statement is this.
- The force of magnitude F acts along the edge of the triangular plate.
- Determine the movement of F about O.
- And there is a force F that is acting, this is the force that is acting.
- What is asked is to determine the movement of F about O, that is this point.
- We can solve it in different ways.

- I can determine what is this angle.
- Similar the that angle will be same as this because this is alternate angles.
- Cos of this angle theta.
- Or I can also take this as a point of application.
- I can resolve this along this axis, which would be a side of this angle theta.
- Then that would into b will also give you give me the.
- Moment about O, or I can simply find out what is this perpendicular distance.
- In whichever way that you want, you can solve it.
- The other way of solving it is I can find out what is this.
- hypotenuse length is in terms of H and B, which is root of H square plus B square, that is what is done here.
- Next we will take up one more problem 2 bar 41.
- The problem statement is A 150 Newton pull T is applied to a card which is wound securely around the inner half of the drum.
- Determine the moment of T about the drum center C.
- Next is at what angle theta should T be applied so that the moment about the contact point is 0.
- The moment about this point should be 0.

Example or application from the lecture

- Now let us take another example from the unsolved problem.
- If I consider the point of application of this force as A.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000138 · EV0000139 · EV0000140 · +1 source units in pack.json

Core explanation · 20:00–24:59

What is that we have to take from this?

Key details

- Then from the diagram I can make out that the inner circle diameter or radius is 125.
- The outer circle radius is 200 millimeters.
- What are we supposed to find?
- First thing is movement about 0.0 because of this.
- Then second thing is at what angle theta we should apply this for such that moment at P is 0.
- What is the moment about.
- If I wound a wire around the circle and pull it, at whatever point that I want to pull, it will always be tangential to the circle.
- This you have to remember.
- It will always be tangential to the circle.
- Force that is P into the inner radius R1, and that's what is written here.
- Next, to find out at what angle theta this should act such that movement about this point is 0.

- If the force simply passes through.
- Or if I if I draw a line such that the line passes through B and also it is tangent to the inner circle.
- Why we should have that?
- Because there is no perpendicular distance.
- If the force passes through any point, the movement produced at that point is zero, that is because there is no perpendicular distance.
- The moment produced at C point.
- And that's what is done here.
- Moment at C is this, that should be equal to what?
- Your P cos theta into R.
- If I equate I'll get this.
- That is one way of solving it.
- The other way of solving it is if I take join this line, this line, this and the tangent.
- This, this, and this, this angle will be at night registration.
- I know this distance is 125, this distance is 200.
- I hope you have understood.
- We will stop today at this.
- We'll continue in our next class.
- That is in class number 9.
- My name is Dr.
- Sventesh from PS University.

Example or application from the lecture

- First thing is what is that we have to take from this given as the magnitude of the force P is 150 newtons.
- So let's take first one.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000142 · EV0000143 · EV0000144 · +1 source units in pack.json

10 UE26ME142A_Unit 1_Class 6_AVSummary

Study focus: UE26ME142A_Unit 1_Class 6_AVSummary. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000146 · EV0000147

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

My name is Dr.

Key details

- Swee Venkadesh from Civil Engineering under Department of Science and Humanities, PS University.
- In the last class we discussed moment and solved a few problems on movement.
- That is in class number eight.
- This is class number nine.
- We continue to solve a few more problems on moment.
- The statement of the problem is in order to raise the flat pole, OC.
- A light frame OAB is attached to the pole and a tension of 3.2 kN is developed in the hosting cable by the power range D.
- Calculate the moment M naught this should be M naught suffix not M naught of this tension about the h point O.
- From the statement, what is that we should think or we should get is the tension in the cable, that is this cable, is 3.2.
- It is tension, so external forces will be moving away, the internal forces will be moving towards that means the force in this cable internal force will be from A towards D and at D it is being from D towards A.
- What would be the movement caused by this force applied onto this light frame AOB at point O because this flat pole when I apply this tension here.
- This has to rotate and stand vertical.
- That's the whole idea of this mechanism.
- Force which is 3.2 kilonewton.
- What are we supposed to find is the moment that oh because of this force that is M naught.
- We are supposed to identify that this OA line is inclined at twenty degrees.
- Similarly, if I draw an horizontal line here passing through A, this angle will also be equal to 20 degrees because these are alternate angles, they will be equal.
- That means it forms an isoscelus triangle.
- In case of an equilateral triangle, the internal angle of the triangle, all of them will be equal.
- That means it will be equal to 180 divided by 3.
- Teta if I take T is the angle made by the internal angle of the triangle that is 180 by 3, which would be 60 degrees.
- For this force, what I need to do is I should take this force, resolve the force along horizontal and along vertical, determine what is the vertical distance from O to A.
- Sine of this angle which is sixty degrees.
- To find out the component of the force, horizontal component, it is 3.2, this angle will be 20, so cos 20, whereas the vertical component will be 3.2 sin 20.
- into the horizontal distance between OA, which is 3 into cos 20, which will produce a clockwise movement, so that's why it is negative.
- So that would be the moment at point O because of the application of the force equal to 3.2 RTA, which is the tensile form.
- Next problem is 2 bar 56.
- The mechanism shown is used to lower disabled persons into a very ball tube or therapeutic.

Example or application from the lecture

- Then the sides of this triangle OAB is given as 3 meters.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000146 · EV0000147 · EV0000148 · +1 source units in pack.json

Core explanation · 05:00–09:59

Treatment in the uploaded configuration, the weight of the boom and hanging chair induces a compressive force of 5 sub 25 newton in hydraulic cylinder A B.

Key details

- Compressive means that the force which cylinder AB exerts on point B is directed from A towards B.
- If theta is equal to 30 degrees, determine the movement of this cylinder force acting on pin B about A point O and B point C.
- Which is used for lowering the disabled person into the will pole.
- Which is a com which is going to be a compressive force.
- He says that if it is compressive, means that the force is exerted from.
- If I from A to B at point B that is this is the force that is the internal force.
- The compressive force means the external forces will be moving towards each other, the internal forces will be moving away.
- We have to find out moment about O and moment about C.
- We can make out the distances from O to A, the vertical distance from O to A, which is 200, from A to C as 800, from C to this point as 175, from this point to this point as 100.
- Then the inclined distance along this point to this point is 400, then from this point to this edge is 900, then very important is this distance from B to this.
- We have a force acting at B.
- First thing is if I draw a vertical line from B till the horizontal line extended from A, it will cut at some point here.
- Now first thing is to find out this angle.
- Why should we find out this angle?
- To find this angle,.
- I should get this what horizontal distance, sorry, vertical distance and then horizontal distance.
- Vertical distance I can find it as 800 plus 175 plus 100 plus this small distance which I can take it as distance from this point to here.
- Which I'll get it from this 400, from this 400 sine of theta, I'll get this angle, this distance, sorry, this distance.
- Then I'll subtract from 150 this because this is 90 degrees.
- The horizontal distance between this also I can determine 400 cos theta I'll get this distance from here to here.
- Or find out 5, if I want to find out 5, which will be equal to 10 inverse of this vertical distance and this horizontal distance.
- What would be this vertical distance?
- 800 as I have told you explained, 800 plus 175 plus this is 400.

Example or application from the lecture

- The magnitude of the force is given as 575 newtons.
- That will be the sense of the force, and he also says if the angle theta that is this angle, the boom angle with respect to the horizontal is 30 degrees, what is given here, then determine what is the movement caused by this force about point O and about point A.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000150 · EV0000151 · EV0000152 · +1 source units in pack.json

Core explanation · 10:00–14:59

If I take sine theta, I get this vertical distance, but I have to subtract this vertical distance from this to this.

Key details

- Cause of this angle theta, this would be theta itself because this line and this line are not 90 degrees.
- If this is theta, if I draw a line here, okay, if this is theta, this will be 90 minus theta and this would be theta.
- That is this distance, 175 plus 400, which is this, cause of this theta, I get it till this point, so 175 plus this, plus I should take this horizontal distance, which is 150 sin theta, but I will be getting from here to this point.
- Once I get this phi, I can resolve this force as 575 cos of phi, I get the.
- Vertical force 575 sine of 5, I get the horizontal force.
- The vertical distance would be this, plus I need to add this 200, or in another way, I can take the point of application of this force as A itself.
- Then this horizontal component is to 200, which will be clockwise.
- I'll take this vertical component into this distance 200, this horizontal component into this distance which is 800.
- 575 sin 5 into 0.2 minus 575 cos 5 into 0.2 we will get the moment about O M naught.
- Moment about C will be 575 sin 5 into 0.2 plus 575 cos 5 into 0.8.
- I get this moment.
- I hope you have understood this.
- Next problem is 5 bar or sorry 2 bar 557.
- The asymmetrical support arrangement is chosen for a pedestrian bridge because conditions at the right end F.
- Do not permit a support tower and anchorage.
- During a test, the tensions in cable 2, 3, and 4 are all adjusted to the same value T.
- If the combined movement of all four cable tensions.
- About point O is to be 0.
- What should be the value T1 of the tension in cable 1?
- That is what is asked.
- Then determine the corresponding value of the compressive force P.
- At O resulting from the four tensions applied at A.
- Neglect the weight of the tower.
- Support, tower, or anchorage.

- Support reaction at B F is zero.
- There is no support here.
- All the supports has to come through the cables.
- The tension in cable two that is this ten, this cable, then three this cable, four that is this cable, all of them have the same tension T, the same tension T in this two, three and four cable.

Example or application from the lecture

- I should not consider this minus 200, only this part, the total.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000154 · EV0000155 · EV0000156 · +1 source units in pack.json

Core explanation · 15:00–19:59

Now for this condition we have to find out what would be the tension in this cable AB, that is cable 1.

Key details

- Then we should also find out what would be the compressive force in this tower AO acting at O.
- We have to neglect the weight of this tower.
- What has to be found also I have explained.
- Next, we should also find out what is the tension in P1 and 10 and the compressive force that is acting on the tower P.
- We should take T as a known quantity that is T.
- Intention T2, T3, and T4 is equal to T.
- Now to find out the movement of O, because he says that the combined movement of all the four forces at point O is 0.
- Then this is an horizontal line.
- Note that the.
- This is not vertical.
- OA is not vertical, it is inclined to vertical at an angle of 5 degrees.
- Theta 2 will be this, which will be equal to.
- The vertical distance which is 15 into the horizontal distance 9 or divided by 9.
- Corresponding to the inclination of the cable 1, 2, 3, and 4 with the horizontal.
- Then T1 is the tension acting which I will take it at D.
- I will resolve all these forces along vertical and horizontal.
- I can find out which will be from this distance and the distance between this.
- I can find out that.
- And that is what is explained here.
- So I've taken T out because that is common.
- Minus T1 cos teta 1.
- into the vertical distance of all of them are 4.8 itself.
- This is the vertical distance.

- All of them will have the same vertical distance 4.8.
- Into I should take this distance from O to B, which will be 6.6.
- That is from the O to A to B, the horizontal distance, then I should also take what is the horizontal distance from A to O, which will be equal to 15 plus 4.8, which is 19.8.

Example or application from the lecture

- The horizontal distance is given as 6.6.
- The vertical distance is given as 15.
- Tower AO is inclined at an angle of 5 degrees, which is given here.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000158 · EV0000159 · EV0000160 · +1 source units in pack.json

Core explanation · 20:00–24:59

And this angle is given us 5 degrees.

Key details

- That is what is entered here.
- Similarly, I will find out for all these three vertical forces acting because of these tensions and the distances.
- But here I have to subtract the horizontal distance from O to A.
- If T is in Newton or whichever units that T is going to be.
- T is the tension in the cable, 2, 3, and 4.
- Next to find out the compressive force in OA, I will take some of all the forces acting along the vertical.
- Note that the.
- Compressive force that is acting in OA will be acting upwards, but that is at inclined at 5 degrees to the vertical.
- That should be equal to all these forces which is acting.
- P1 and this, all this which should be acting downwards.
- Attention in the table two, three, and four.
- I hope all of you have understood today's class.
- We will continue in the next class.
- This is class number nine.
- And I am Dr.
- SV Venkatesh from PS University.
- We will continue in the next class.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000162 · EV0000163 · EV0000164 · +1 source units in pack.json

11 UE26ME142A_Unit 1_Class 8_AVSummary

Study focus: UE26ME142A_Unit 1_Class 8_AVSummary. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000166 · EV0000167

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

My name is Dr.

Key details

- SV Vengitesh from Civil Engineering and the Department of Science and Humanities, PS University.
- This is class number nine.
- In the last class, we had discussed about.
- The problems how we can solve under the topic moment.
- Today we will take up the next topic, which is couple.
- Two unlike parallel forces constitute a couple.
- When I say two, equal and unlike parallel forces constituble.
- When I say to equal, what does it mean?
- Means that the magnitude of the forces are same.
- That is what it means by equal.
- When I say unlike parallel, unlike parallel means.
- The sense of the two forces are opposite to each other, that is unlike.
- Then parallel, they are acting parallel to each other.
- One is moving into the screen, the other one is coming out of the screen.
- They are parallel.
- The magnitude of these two forces are the same.
- So a couple is defined as two equal, unlike parallel forces constitute a couple.
- This is one force which is the center of the forces coming like this.
- This is another force, the center of the force is going with this.
- But the magnitude of the force is F, the magnitude of this force is also F.
- The effect of a couple is to make the body rotate.
- No, how do you say that?
- No, since the sense of this force is like this, the sense of this force is like this, the combination of these two, when I sum it up, it will give me zero because the senses are zero, the sense are opposite.
- The magnitude of the force is the same, so effectively the net force acting is zero.
- But since they are acting at different.
- Points separated by some distance, it would make the body to rotate.
- The rotation can happen in the counterclockwise okay, or it can happen in the clockwise direction, depending upon how the sense of the forces are acting.

- if this forces to be acting along X and this to be acting along Y, then the Z will be coming out of the plane.
- The positive force means it is counterclockwise.
- It is counterclockwise.
- And vice versa, if it is going to rotate in the clockwise direction, it is anti-clockwise.
- It is not translatory motion, but it is rotatory motion.
- Whereas this force is acting at a distance of A plus D.
- The distance between the two forces is D.
- Whereas this force into this total distance A plus D would produce an anti-clockwise movement.
- So the movement produced at O because of this force is this F or it is R B into F in vector approach or R B cross F I should take, then it is plus RA cross.

Example or application from the lecture

- We will take this example and see what would be the movement that is going to be produced at point O.
- Let's take the perpendicular distance of this force from A to B R A, and perpendicular distance of this force from O to be R B.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000166 · EV0000167 · EV0000168 · +1 source units in pack.json

Core explanation · 05:00–09:59

If I want to find out the movement of these two forces, the sense of this would be acting upwards.

Key details

- If I take the take these two forces to be acting in the horizontal plane, the vertical axis perpendicular to the horizontal plane.
- Will be the axis about which the movement is going to be produced, and you can see that this would produce a positive movement that is anti-clockwise movement.
- Also, I have explained.
- What is the effect of a couple?
- That is because the momentum produced by the couple at any point that you take would be the same.
- What is the effect of a couple?
- Then what is the classification of couple?
- And how do we determine the moment produced by a couple at any point, which should be equal to the.
- Distance between the two forces that is acting or constituting a couple into the magnitude of the force that is acting.
- There is a force F, negative force F acting at this point, that means the sense are different.
- They are differentiated or the distance between the two is D.
- The same force.
- F minus F and F, instead of applying it here, this is the point of application of these two.

- Whereas if I change it at this point, it would still be equal to M only.
- Change the position to be here and here, provided this distance is still D, again I will get the moment to be equal to M itself, and that is positive.
- You can see here this distance is half of this distance.
- So this is what is equivalent couple means.
- I already explained what is the characteristics, one of the characteristics.
- The net force acting because of a couple is zero.
- Because one force is acting in the positive direction, the other force is acting in the negative direction.
- So combination of these two will give me zero as the magnitude of the couple, or the force acting because of a couple is zero.
- The movement produced by a couple at any point is same.
- I have explained this also.
- The moment produced by a couple is same at any point.
- Or the movement produced by the couple at any point is same.
- Next, a couple.
- How do I balance a couple?
- I can balance a couple by an equal and opposite couple only.
- I cannot balance it by a single force because the net force acting is zero.
- A couple cannot be replaced by a single force.
- That is also a characteristics.
- These are the characteristics of a couple.
- What does this mean?
- The magnitude of the force is F, it is acting along this line of action, and the sense is moving away from A towards the right side.
- At B, what is the condition?
- The condition is that whatever effect is produced by this force acting at A, the same should be the effect produced by the system of force that's I am going to act at B.
- A that is the condition.
- That is what is called as a force couple system.

Example or application from the lecture

- So a couple is an example for a free vector.
- For this, I'm taking a simple example of a force acting at a point A.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000170 · EV0000171 · EV0000172 · +1 source units in pack.json

Core explanation · 10:00–14:59

So to do this, first thing what has to be done is I have to remove this force.

Key details

- To remove this force, what I'll do is I'll apply an equal and opposite force at B.
- The net effect of these two would be zero.

- But I cannot apply simply at P this force because it will change the.
- Effect of a force.
- I get only this again.
- But now I'll take this force minus F and F acting at A.
- Those two are now constituting a couple.
- I'll remove that and replace it by a couple, that is moment M, which is equal to F into D.
- We know that the moment produced by a couple is.
- Force into the distance between the two forces constituting a couple.
- The effect of this would be the same as the effect of this.
- Because this will have the same effect as this, because these two will cancel with each other.
- I will get only this.
- How I have replaced these two forces that is minus F and F.
- acting at A by a moment so this effect and this effect is same so eventually this effect and this effect are also same this is what is called as a force couple system and we will be using exclusively this concept for solving problems.
- I'll read out the statement of the problem.
- This is a solid problem in the textbook.
- The rigid structural member is subjected to a couple.
- Consisting of the 200 newton forces replace this couple by an equivalent couple constituting of the two forces P and P dash, each of which has a magnitude of 400 newton.
- Determine the proper angle T tar.
- If you refer to the diagram, you will be able to know.
- The point of application of the two forces of magnitude 100 newtons.
- One is positive, the other one is negative.
- P and minus P acting at these points at the top of the diagram.
- But the forces P is acting at an angle of theta with respect to this line.
- The effect of the forces 100 newtons, which is separated by 100 millimeter is going to produce a couple positive couple of magnitude 100 into 100.
- Because we have to replace that by this.
- along this line which will be equal to p cos theta.
- When I resolve this along the vertical.
- Which will be P sine theta, and this one will be P sin T again.
- They are acting in the same line and opposite to each other, so they will balance with each other.
- Whereas this and this, they are not acting in the same line, but they are separated by the distance 40 mm.
- Then get the value of P.
- From that equation I will be able to get the value of P.

Example or application from the lecture

- Next, let's take this simple example.
- They are separated by a distance of 100 millimeters because here it is given dimensions in millimeters.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000174 · EV0000175 · EV0000176 · +1 source units in pack.json

Core explanation · 15:00–19:59

Next, let's take this example.

Key details

- By a force couple system acting at O.
- If I want to replace this force by force couple system at O, what is to be done?
- Just apply the 400 newton force, whatever is there acting at this point, bring it to this point, which will be 400 newtons acting at O like this, then determine.
- What is the movement produced by this force at O?
- How do we find that?
- We take this force, multiply it with the vertical distance between O and this point.
- Which would be equal to 200 sine of 60 degrees, which would be the vertical distance.
- The force couple system acting at O will be 400 newtons acting at O to the left, then a counterclockwise movement which will be equal to the magnitude of 400 into 200 sin 60.
- That is what is explained here.
- For this force, I will apply an equal and opposite force here.
- So that means simply bring this force here.
- Replace them by a moment.
- Which is equal to four hundred, that is this, into two hundred is this distance, sine sixty because this angle is sixty, the vertical distance is sine sixty.
- We have explained in this today's class, that is class number nine, what is a couple?
- What are the characteristics of a couple?
- What is the effect produced by a couple?
- Then how do we determine the moment produced by a couple?
- Then we have looked at what do we mean by force couple system.
- How do we get it?
- A force couple system.
- How do we solve problems using the force couple system?
- I hope all of you have understood.
- This is class number nine.
- We will continue to solve some more problems from the unsolved.
- Okay, in our next class that is class number 11.
- We will continue in the next class that is class number 11.
- My name is Dr.
- She Vingtej from Civil Engineering under Department of Science and Humanities, PS University.
- See you in the next class.

Example or application from the lecture

- The force coupled system is nothing but replacing a given force acting at a point A by some system acting at point B by a force couple system such that it produces the same effect, and

we also have looked at.
- Okay, two problems, example problems I have taken and I have explained these two problems.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000178 · EV0000179 · EV0000180 · +1 source units in pack.json

12 UE26ME142A_Unit 1_Class 9_AVSummary

Study focus: UE26ME142A_Unit 1_Class 9_AVSummary. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000182 · EV0000183

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

My name is Dr.

Key details

- SV Venkitish, Civil Engineering Department of Science and Humanities, PS University.
- This is class number 11.
- In the last class we discussed about couple, the effects of couple, the classification of couple, the characteristics of couple, force couple system, then we solved a few problems.
- Today, that is in class number eleven, we will try to solve few more unsolved problems from the prescribed textbook that is Engineer Mechanics Tactics by Miriam, Craig and Balden.
- Use in this problem, which is 2 bar 68, the statement of the problem is the force F is applied at the end of ACD, which is mounted to a vertical post.
- Replace this single force F by an equivalent force couple system at B.
- Next redistribute this force and couple by replacing it with two forces.
- acting in the same direction as F, one at C and the other at D, and determine the forces supported by the two X bolts.
- Using values of F equals 425 newtons, theta equals 30 degrees, B equals 1.9 meters, D equals 0.2 meters.
- H equals point eight meters and I equals two point seven five meters.
- There is a force that is acting at the end of the arm A C D, that is at this point A.
- We have to replace this force by an equivalent force couple system at B, that is at this point.
- Redistribute this force and couple by replacing it with two forces acting in the same direction as F.
- Same direction of F means it is at this angle theta.
- He has not specified the sense of the force.
- He has specified that the angle of the force that is going to be acting one at C and one at B should have the same direction, which means it is at angle T with respect to the vertical.
- and determine the forces supported by the two X bolts.

- X boles is these are the two bolts that is C and D.
- What are the forces acting at that?
- Then the angle theta is 30 degrees.
- Now, first thing is we have to replace this force by a force couple system at B.
- I will find out the component of this force and use that as a force that is going to be acting at B.
- P is equal to the component of this force here along X would be to the left.
- I am taking.
- X axis to the right as positive and upwards y axis as positive.
- Then component of this force along Y will be downwards, so again negative.
- and it's going to be a cause of this angle theta j.

Example or application from the lecture

- And then he has given you the values of the force F which is 425 newtons.
- Then the B distance that is from A to the center between the two volts that is C and D is given as 1.9 meters.
- Then the distance D that is from the center of the bolts C and D till the C, R T is given as D and D.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000182 · EV0000183 · EV0000184 · +1 source units in pack.json

Core explanation · 05:00–09:59

Minus three sixty eight point zero six J Kilo Newtons because sorry, if this is in Newtons, I'm sorry this has to be changed to newtons because what is given here is in newtons.

Key details

- So once we find that, the next thing that we have to do is we have to determine the moment about B because of this force that is acting.
- So moment about P because of the horizontal component will be F sine theta into L minus H because the horizontal component vertical distance is L minus H.
- Then we will take the vertical component which is F cos teta into the horizontal distance is V.
- Straight away we can put that, then substitute the value for F, L, H and B, we get the moment about B.
- which is equal to F sine theta i with a negative sign minus f cos theta j which is acting along the y axis that is the force then movement we have already got because of the two components and multiplying it with the respective distances perpendicular distances.
- The second part of the problem is to replace this force and the couple by two forces acting at C and D, which will have the same direction, that means this direction.
- The movement acting at P will be in the anti-clockwise direction.
- That should produce a movement at B in the anticlockwise direction.
- Then when I apply the force at D, that should also produce a movement at B in the anticlockwise direction.

- This force when I apply, this force should produce a moment at B which is in the anti-clockwise direction.
- That would be appropriate as what I feel.
- Or F B and also the movement at B, whatever is there, should be the movement that is going to be produced by these two forces.
- To be equal to 0.
- So we will get this force F C which is acting downwards minus F C which is acting upwards should be equal to F because this is the total force that should be.
- That we should get when we sum up these two, so that I have equated, and f we know that it is equal to 425 newtons.
- No momentum or B whatever we get because of these two.
- That would produce an anti-clockwise movement, so positive.
- Then the component of F C in the horizontal direction, that is F C sin theta into the vertical distance from B to this point B to C, which is L minus H, and that will also produce an anti-clock movement, so positive.
- Next, the component of the force along vertical, which is because of Fd, would be Fd.
- cos teta into D into D which will be positive because that will also produce an anti-clockwise movement.
- But the component of F T acting in the horizontal direction that is going to be this would be the sense that will present clockwise movement.
- Sine theta into the horizontal vertical distance is L minus H, so that is what is indicated here.

Example or application from the lecture

- Okay, that's the first part of the problem to replace the given force which is acting at A point to a force couple system acting at B.
- We are taking so that would be equal to F C, which is this component of this acting along the vertical, which is F C cos theta into D, that is what is given here.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000186 · EV0000187 · EV0000188 · +1 source units in pack.json

Core explanation · 10:00–14:59

Then theta is 30 degrees and D we all know it is 0.2.

Key details

- From this we would be able to get the value of F D which is equal eight hundred and seven newtons.
- I hope all of you have understood.
- You have to replace that by a force couple system at B.
- Then replace this force couple system by.
- Forces acting at C and D is what we have done.
- Next, another problem we will discuss, that is problem number 2 bar 76.
- Seal seed back release mechanism.

- The part is subjected to the four newton force exerted at A and a 300 newton millimeter restoring moment exerted by a hidden torsional spring.
- Determine the y-intercept of the line of action of the single equivalent force.
- And there is a couple or a moment of 300 newton millimeters that is acting.
- We have to replace this by a single equivalent force.
- And we should find out where this single equivalent force should act along the y-axis so that produces the same effect as that of the single force of and the moment 300 newton millimeters.
- This one, this problem is.
- Opposite of that of the force couple system.
- Then we can also note that the point A is at a distance of 10 millimeters from O horizontally.
- and it is at a distance of 40 millimeters from O vertically.
- What is that single equivalent force that should be applied on the y axis so that it produces the same effect.
- 40 newtons, but we have to apply this force on the y-axis such that force when I apply it on the y-axis would produce a movement which will be the combination of the movement produced by F acting at A and this movement 300 newtons.
- So, first thing that we need to find is what would be the combined moment by this force F and the moment 300 newtons that would be acting.
- So first thing is what would be the moment M naught.

Example or application from the lecture

- What we used to do earlier was there was one force given, we used to determine what should be the force couple system, which would produce the same effect when it is applied at some other point, is what we used to determine.
- Going forward, note down what is given, the force F is given, the moment is given.
- From the diagram, we can note down that the angle at which the force is acting with respect to the horizontal axis is 15 degrees, that is also given.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000190 · EV0000191 · EV0000192 · +1 source units in pack.json

Core explanation · 15:00–19:59

Combination of this and this is what we have to determine.

Key details

- Sine theta by theta is 15 degrees, so this component we will get into this distance 10.
- So we will get the moment because of this component along y.
- So movement because of the component along X would be in the clockwise direction, so it is negative.
- The component will be of cos theta into the vertical distance which is 40 mm.
- Only the component at the horizontal component would produce a moment.
- Cos 15 into y should be equal to the moment at M naught.

- I hope all of you have understood.
- The two problems that I have discussed today, which are not solved in the textbook, I have exclusively taken these two problems, which are.
- Tougher and not easy to understand.
- I hope all of you have got this problems.
- This is class number 11.
- We will be continuing the class.
- My name is Dr.
- I am from Civil Engineering under the Department of Science and Humanities, which comes under PS University.
- See you in the next class.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000194 · EV0000195 · EV0000196 · +1 source units in pack.json

13 UE26ME142A_Unit 1_Class 10.0_AVSummary

Study focus: UE26ME142A_Unit 1_Class 10.0_AVSummary. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000198 · EV0000199

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

I am Dr.

Key details

- Svenkitesh from Civil Engineering Department of Science and Humanities, PS University.
- In the last class, we discussed a few problems on couple.
- Today we will be discussing more problems.
- The statement of the problem is an overhead view of the handlebars on an all-term vehicle is shown.
- If the indicated forces have a magnitude of F equals to 150 newton, determine the moment created by the two forces about the vertical steering axis through point O.
- Both n axis are perpendicular to the left handlebar.
- Treat the problem as two dimensional.
- What is asked is to determine the moment created by these two forces about the vertical steering axis through point O.
- Point O is here, this is the vertical steering axis.
- The axis is perpendicular to the screen.
- The vertical axis passing through O is perpendicular to the screen.
- The sense of the force acting at A is towards A, it's going towards A.
- Whereas the sense of the force F at B is coming out of B.

- How can I make out that?
- He says that the N axis, both these axis, M.
- or perpendicular to this handle axis.
- He says very clearly both n axis are perpendicular to the left handlebar.
- Both n axis are perpendicular to this handle axis.
- Here also it is at 10 degrees.
- F and F is the magnitude, and it is 150 newton.
- The magnitude are same, they are parallel to each other, the sense is opposite.
- That means what?
- It is a couple.
- A couple is.
- Defined as when two equal unlike parallel forces are that is going to constitute a couple.
- We have two equal, unlike parallel forces, so they constitute a couple.
- That's a property of a couple.
- Next, if you look at this, it says OA is equal to OB.
- That means the horizontal distance of O to A and O to B are equal and.

Example or application from the lecture

- The magnitude of which is given in the statement as one fifty newtons.
- That is why the distance which we need to be, the horizontal distance we need to be, is given as 750 mm.
- They were also given this angle as 15 degrees.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000198 · EV0000199 · EV0000200 · +1 source units in pack.json

Core explanation · 05:00–09:59

We need not have to use this 15 degrees at all.

Key details

- Y means if I could get the component of this force along the vertical, which is equal to how much?
- This axis and this is horizontal.
- This is a vertical axis.
- The horizontal and vertical will always be at 90 degrees.
- will be ninety minus fifteen, which will be seventy-five degrees.
- Next we also have y.
- And the axis of the handle, which is at 9 to each other.
- Width of vertical will be equal to this 10 degrees plus this 50 degrees, which will be 25 degrees.
- This would be moving upwards, whereas this component will be moving downwards.
- What happens to the horizontal component?
- The horizontal component of F would be F sine 15.
- Sign 15 degrees.

- That will not create any movement.
- If I want to find out the moment, I can take only the vertical forces, take moment about A.
- We'll give the moment at A.
- That moment, whatever we are going to get, would be the moment at any point you take it.
- Okay, so the movement at O will also be equal to the movement that we get because of these two forces about A, or the movement that we get because of these two forces about B.
- I hope it is clear.
- When I take this force, this is fifteen, this is ten, so this would be twenty-five, so I take it as F.
- Cost twenty-five, multiply that with this distance that is 0.75.
- This is in M, so converting it into meters.
- Whether I take it at A or at B or at O, it is going to be same.
- 2.0 newton meters and it is gonna be clockwise.
- No, you can also solve it by considering the point O, resolving this force along vertical and horizontal.
- Taking moment about all because of it, because of those components, which you will get it as 150 cos 25 into 750 by 2.
- This is the horizontal distance.
- Then you can resolve this.
- That would be 150 sin 25 into the vertical distance would be 750 divided by 2, 10 15 degrees, which you will get here as the.
- This would also produce a clockwise moment.
- The magnitude of that would be 150 cos 25.
- into 750 by 2, then we will get this component which will also produce an anticlock as point.
- The property of a couple is that moment about any point because of the couple is going to be same.
- That is what we are going to be using in this equation, and you can get it very simple.
- one line answer for this problem even though it looks when you look at the diagram it looks very complicated but if you have understood clearly the properties of a couple then it becomes very simple okay right.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000202 · EV0000203 · EV0000204 · +1 source units in pack.json

Core explanation · 10:00–14:59

Let's look at the next problem, which is 2 bar 74.

Key details

- The statement of the problem is the force F is applied to the leg extension exercise machine as shown.
- Determine the equivalent force couple system at point O.
- Use values of F is equal to 520 newtons, B equals 450 mm, H equals 2 point.
- 215 mm, H is equal to 215 mm, R is equal to 325 mm, theta is equal to 15 degrees, and phi is equal to 10 degrees.

- Then what are the distances at which the force is applied with respect to the point where we want to find out the.
- Moment or the couple.
- The horizontal and the vertical distance or the distance along this line between point O and A, and the vertical distance perpendicular to this line is what we have to look at.
- Or rather, I can take it as R minus H, I will get this distance that is from A to O.
- You take this force, you have the axis, this axis, okay I can draw the same axis here, apply this force at an angle of theta with respect to this, that is this axis, so this is theta, or I can resolve this force.
- Yes, cos theta would be the component of the force along this axis, so I can apply that.
- Then I can resolve this force along this, that is the line which is parallel to this.
- There is this one which will be F sine theta.
- That will be the four that is acting at O.
- Then the component of this force acting at A, provided R is greater than H would be F sine theta into R minus H.
- This would be acting in the anticlopped direction, so it is going to be negative.
- This is a free body diagram.
- We have R, this is angle theta, then this is a point A, this is O R so the.
- Same diagram whatever we have looked at it, the free body diagram is written here.

Example or application from the lecture

- Let us worry about only drawing the free body diagram from that.
- Very simple, they have given this as R.
- Then this angle phi is given as 10 degrees.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000206 · EV0000207 · EV0000208 · +1 source units in pack.json

Core explanation · 15:00–19:59

The angle made by this would be 10 plus 15 is 25 plus 90 is 115 degrees.

Key details

- However, you can take that into consideration or you can solve it in the way that I have told.
- which is this force, the component along this into B plus R, as I've told you, B plus R is the perpendicular distance.
- That would produce a clockwise moment which he has taken as positive.
- Then we have this force, this component F sin theta, which is this.
- Then into R minus H which will produce an anti-clockwise movement, he has taken it as negative.
- We can take clockwise as positive, anticlockwise as negative, or clockwise as positive, anti-clockwise as negative, vice versa also is true.
- In this class, we have solved two problems.
- That is problem 2 bar 73 and problem 2 bar 74.
- In the problem 2 bar 73, we have used the properties of the couple, which indicates that.

- The moment produced by a couple about any point is same.
- In this case, we have used the concept of force couple system.
- Counterclockwise above horizontal.
- Where F is moving from O to.
- This line if I extend this angle will be 10 plus this angle perpendicular to this line will be 90.
- In the counterclockwise direction from the positive x axis okay I hope everybody has understood these two problems.
- In the second chapter, we have discussed so far three topics.
- First one was the resolution of forces.
- Where we have looked at the rectangular components, and what is the difference between projection and component, and how to get the component of the force if the two axis are perpendicular to each other, and where.
- The two axis are not perpendicular to each other.
- In the second topic, we have discussed about couple.
- In the third topic, we have discussed about couple.
- We'll see you in the next class.
- My name is Dr.
- From Civil Engineering in the Department of Science and Humanities, PS University.

Example or application from the lecture

- Let's meet in the next class.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000210 · EV0000211 · EV0000212 · +1 source units in pack.json

14 UE26ME142A_Unit 1_Class 10.1_AVSummary

Study focus: UE26ME142A_Unit 1_Class 10.1_AVSummary. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000214 · EV0000215

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

My name is Dr.

Key details

- Sviven Kitish, Civil Engineering, Department of Science and Romantics, PS University.
- In the last class, we had discussed.
- few problems the unsolved problems from the prescribed textbook, Engineer Mechanics Statics by Miriam, Craig and Holton.
- Today is the thirteenth class, we'll be continuing to solve few more problems on the topic couple.

- The statement of the problem goes something like this.
- The system consisting of the bar OA.
- Two identical pulleys and a section of thin tape is subjected to the two 180 newton tensile forces shown in the figure, determine the equivalent force couple system at point O.
- There is a force of hundred and eighty newton force which is tensile force applied on a bar OA.
- This is bar OA.
- There are two pulleys, one is here, the other one is here.
- The force of 180 newton tensile is applied on these two pulleys.
- Which are both of them are horizontal, then he has asked us to determine the equivalent force couple system at point O.
- So equivalent force couple system at point O would be comprising of the net force because of these two forces plus the movement produced by these two forces at this point.
- Both of them are horizontal.
- The upper force is acting to the left, the lower force is acting to the right.
- You can look at that they form two equal unlike parallel force system, which is nothing but a couple.
- From the characteristics of a couple or properties of a couple that the net force acting because of a couple is zero.
- And that's what we will get here also.
- 180 newton force is acting to the left.
- The upper one, 180 newton force, the lower one is acting to the right.
- So to find out the force couple system at O, because of these two forces, there won't be any force.
- About if I take this force, I'll take the moment about this point.
- If I take this force, I'll take the moment about this point.
- That would be the moment at or at the center or at a point or anywhere you take, that would be the moment that is produced.
- Because it's a couple.
- This is horizontal, so I would like to know what is the vertical distance between the two forces, which is 180 and 180, these are the two forces.
- How do I get the vertical distance between these two forces?
- And we also know that OA is inclined at 45 degrees.

Example or application from the lecture

- Which has two identical pulleys, and if you observe the diagram that is given from this diagram, you can note on that.
- Then they have also given you what is the distance from O to the center of the lower pulley along the line OA, which is 50 mm, then the distance between the centers of the two pulleys along OA, which is 100 mm.
- He is also given the radius of the pulley as R and the value of R is also given as 25.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000214 · EV0000215 · EV0000216 · +1 source units in pack.json

Core explanation · 05:00–09:59

Sine 45 would get the vertical distance between the two centers of this pulley.

Key details

- We also know that this is radius, which is 25.
- Plus 25 will be the vertical distance.
- Which is acting in the anti-clockwise direction, which should be equal to the magnitude 180 newtons, cross or into the vertical distance between the two forces.
- And you can see the same thing here.
- This is the radius.
- This is also the radius.
- So obviously this would be the radius because this is the tangent, this would be acting tangent to this circle.
- You will get this as 21700 newton millimeters or 21.7 newton meters, which will be acting in the anti-clockwise direction, and we take anti-clockwise direction as positive.
- I hope all of you have understood this, right?
- We resort the force along the horizontal and multiply with the distance to get the movement at the pivotal point of the handle that is an O.
- Okay, because they are also a couple.
- Vertical distance, which is the perpendicular distance to these forces.
- I hope all of you have understood this, right?
- I think of this problem.
- This is a very important problem, and this problem.
- Which is 2 bar 78, the statement goes something like this.
- The force F acts along line MA, where M is the midpoint of the radius along the x-axis.
- Determine the equivalent force couple system at O if theta is equal to 40 degrees.
- From the statement, what we can know is there is a force that is acting.
- The line of action of the force is MA.
- The point M is the midpoint of the radius along the x axis, that is this is the x axis.
- Then they have also says that the angle theta is what it is.
- What is this angle theta?
- This is angle theta, which the radius OA makes with the x-axis.
- The force couple system of this force which is acting at this point, A, along M, a force couple system acting at so how do we solve it?
- Now I have to resolve this force first along horizontal and vertical.
- To resolve this force along horizontal and vertical, I should get this angle, say alpha.
- To get this angle alpha, this problem is majority of geometry and very less mechanics part is there.

Example or application from the lecture

- Who forces are acting parallel to each other on the sides or the edges of the handle of the two forces, there were acting parallel and horizontal distance was given.

- This is the radius and this M is the midpoint of this radius, and they have also given that the radius is R or they have given here this distance is R by 2, this is also R by 2, so the radius would be capital R.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000218 · EV0000219 · EV0000220 · +1 source units in pack.json

Core explanation · 10:00–14:59

I can find out what is the horizontal distance OB and the vertical distance AB.

Key details

- Because I know that this is R and this angle is theta.
- AB by M B.
- Or I can apply this force F here, which makes an angle of alpha with the x-axis which we have found.
- Next to determine the moment about O, I can take this force which is a component of F along x axis, which is F.
- Cos alpha multiplied with AB, so which will produce a clockwise moment which will be negative.
- The component of this force along y will be F sin alpha into this distance which is OB, which we have already found, which will produce an anti-clockwise moment which will be positive.
- This is just the arc.
- Then we have OA.
- Then we have.
- Now as I told you this angle is considered as alpha, which will be the same as this, because they are corresponding angles.
- I can get this to be equal to.
- I you can solve it by the approach that I have explained earlier, or by this approach, you take this triangle and use.
- The cosine rule.
- What does the cosine rule state?
- The side, any side, if I want to find out, a m that is this distance, a m square is equal to.
- This the other side R square and one more side plus R by 2 square minus 2 times R into R by 2 the other two sides.
- Cause of the angle opposite to the side on which we are we are supposed to find out that is AM.
- Opposite to AM is theta, that was that is what is mentioned here.
- I can find out alpha as 180 minus beta.
- This is another way of solving or determining the angle alpha.
- I have explained two different ways of solving alpha.
- You can use n number of ways to solve alpha.
- Determining of alpha is left to your knowledge that you have gained till now.
- You can solve it in n number of ways to determine alpha.

- Now once I get alpha by whatever means and method that I get, but.
- Make sure that the angle that you get is correct one, 67.5.
- Use correct logic to determine the angle alpha.
- Once you get angle alpha, now I will find out what is the moment about O by resolving this force along horizontal and vertical.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000222 · EV0000223 · EV0000224 · +1 source units in pack.json

Core explanation · 15:00–19:59

If I take this force to be acting at this point M itself, I can take this as F.

Key details

- Sin alpha into r by 2, that's what is taken here, which produces an anti-clockwise moment.
- Then I'm taking this as a point of application, so this component will be F cos alpha, F cos alpha, or I can also take the component of this by taking.
- Instead of alpha I'll take.
- Alpha is 65.
- M naught is equal to F sin alpha.
- This component, whatever you get cos alpha into there is no perpendicular distance because this four this component will pass through O itself.
- Mind it, what he has taken is F sin alpha is sine 67.5.
- He has not considered this component because it passes through O itself.
- Magnitude of F is same as F, the sense of the force will be the same as this, which is moving away from O or moving upwards.
- Then the direction of this force with respect to the x axis measured in the anti-clomage action positive x axis would be 67.5 plus the movement that's going to be acting at O will be equal to the value whatever we get from here which is 0.462 F into R.
- Earlier, what I explained was considering this as a point of application, I had explained how to get the moment about O.
- I had also explained how to get this angle alpha by taking or constructing a triangle.
- a b m and determining the distance mb and ab and then finding out alpha once you get alpha you take the point of application of this force as a itself determine the movement about O by resolving the forces of F along horizontal and vertical.
- Both the ways you can solve it.
- Of course, you can use n number of ways to solve this problem.
- That is, you can also resolve this force along this line.
- OA provided you can find out what is the angle made by the force of with respect to OA, and then resolving this along O and perpendicular to O A, then you will get F.
- component along this axis which is perpendicular to OA into this distance R, you will still get this answer provided you find out that angle whatever F makes with the line OA.
- And that is what we have done.
- The topic couple.

- This is the third topic under unit one, chapter two, we have discussed first topic was on resolution, rectangular, then projection, then N and T components.
- Then what is the difference between projection and component?
- Then the second topic was on moment.
- We have discussed how to determine moment, what is the positive and negative sign of moment.
- The third topic was couple.
- We have discussed couple characteristics of couple.
- The sense and classification of couples.
- Then we have discussed what is force couple system, and we have solved few problems.
- from the solid examples and few problems on from unsolved examples.
- In the next class that is class number fourteen, I'll start with the last topic of unit one that is resulted.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000226 · EV0000227 · EV0000228 · +1 source units in pack.json

Core explanation · 20:00–24:59

We have already discussed few things about resultant.

Key details

- I will discuss more about resultant in my next class.
- That is class number 14.
- My name is Dr.
- SV Venkage from the Department of Civil Engineering under Science and Humanities Department, PS University.
- See you in the next class.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000230 · EV0000231 · EV0000232 · +1 source units in pack.json

15 UE26ME142A_Unit 1_Class 11_AVSummary

Study focus: UE26ME142A_Unit 1_Class 11_AVSummary. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000234 · EV0000235

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

I am Dr.

Key details

- Svinkitesh from Civil Engineering under Department of Science and Humanities BS University.
- In the last class, we had discussed about the solving of problems on couples, and I had also informed you that in the next class, that is class number fourteen, I would be starting the last topic of unit one, that is resultant.
- What is resultant?
- Resultant, I had also mentioned you regarding resultant in the very few first classes.
- Resultant is that single force which would produce the same effect as that of us system of forces.
- If you apply the system of forces on a physique body, there could be some effect.
- The effect could be a translatory motion or a rotatory motion or the combination of these two.
- It can be translatory as well as rotatory.
- This single force would also produce the same effect as that of that system of forces, then this single force would be called as the resultant.
- I talked about dot product and when I talked about dot product I told you if you have vectors or if you represent the forces as vectors and get the dot product of this vectors, that would give you the resultant.
- That means it would give you the vector form of the resultant.
- That means it will give you the magnitude of the component of the resultant along x axis, y axis.
- It would also give you the sense of the components of the resultant along X along y.
- Then it would also tell you what is the direction of these components, that is, it is along X and along Y.
- You should also take the sign of the forces or the vector into consideration.
- This f sigma f x will be in the form of Fx i or RX i plus R y j or Rx I minus R y j.
- What about the sign convention according to the forces that is acting.
- The sum of all the forces or the component of all the forces acting along X, algebraic sum of all the components of the forces acting along X, which should be nothing but your Rx.
- Components of the forces acting along y, which will be r y, and then I can find out the magnitude of the resultant R is equal to root of summation of R square plus summation of R y square.
- Would give us the magnitude of the resultant.
- The theta will give us the direction of the resultant with respect to the x-axis.
- Based on whether this is positive or negative, I would get the sense of the component of the resultant along x axis positive or negative, or y axis positive or negative.
- When we club these two, I get the sense of the resultant.
- The same thing is explained here.
- Say if you have a rigid body.
- And there are four three forces acting F1, F2, and F3.
- Now, because of these two forces F2 and F3, I can find out the resultant as R1 using the law of parallelogram of forces.

Example or application from the lecture

- Let us look at this.
- For example, you have a system of forces.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000234 · EV0000235 · EV0000236 · +1 source units in pack.json

Core explanation · 05:00–09:59

Adjacent sides of the parallel and then find out the resultant R.

Key details

- This is one way of finding out the resultant.
- The other way of finding out the resultant is explained here.
- Each force you find out the component of the forces.
- X and Y similarly for F2 again F2x and F2Y for F3 also again you find out F3x and F3Y.
- F3x is also positive, whereas F1 y is positive.
- F2y is also positive, whereas F3Y is negative.
- Adding all the components.
- Algebraically I get R y.
- We know that for a force there will be four characteristics.
- What are these four characteristics?
- I had explained this earlier also, I will repeat it.
- First one is the magnitude of the force.
- Or the quantity of the force is the magnitude.
- The second one is the direction, which is the angle made by the force with respect to a reference axis.
- The sense or line of action of the force, whether the force is moving away from the point of application or moving towards the point of application, and the fourth one is the point of application.
- If all the three forces have the same point of application, or if it is a.
- Concurrent force system, the resultant will also have the same point of application.
- But if it is not a concurrent force system, as you can see here, this is one force, this is another force, this is another force.
- When we produce these three forces, all the three of them will not meet at the same point.
- In such case, where am I supposed to apply the resultant force is the question.
- To find out the point of application of the resultant, we go for what is called as principle of movement, or it is also called as extension of varignos theorem.
- Or it is also called as law of movements.
- I would define all this.
- If you remember I defined this when I was talking about the second topic in unit one, chapter two, that is movement.
- What does Varing North theorem state?
- Varing North theorem states that if you have a force, the moment produced by the force would be equal to the moment produced by its force components about the same point.
- Very importantly, about the same point is very important.

- The moment produced by a force.
- About a point is equal to the moment produced by the component of the force about the same point.
- Now coming to principle of moments or law of moments or extension of Varingon's theorem means.
- The moment produced by a system of forces about any point will be the same as the moment produced by their resultant about the same point.
- I'll repeat.
- The moment produced by a system of forces about any point.
- Is equal to the moment produced by their resultant about the same point.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000238 · EV0000239 · EV0000240 · +1 source units in pack.json

Core explanation · 10:00–14:59

And they have mentioned the distances D1, D2, and D3, which are the perpendicular distance of this force F1 and F2 and F3 from point O.

Key details

- That means this angle is 90 degrees, this angle is also 90 degrees, this angle is also 90 degrees.
- Along with this, I will also apply this moment M1, M2, and M3, which is nothing but the force couple system.
- If you remember, so instead of applying this force at this point A, F1, F2, and F3, I can apply these forces at O by a force couple system, and that's what is indicated here.
- The sense of that.
- If I take this about this point the moment will be anti-clockwise and that is what is indicated here.
- Then if I take this force, the moment about this point will be this force into this distance, which will again be clockwise, and that is what is indicated here as M3.
- That is equal to summation F into D.
- Then you add all these three, that is dot product, you get R, which is sigma F.
- Somewhere away from O at a distance perpendicular to this R, that is D.
- The four characteristics.
- One is the magnitude, which is the dot product of all the forces acting on the body.
- Then how do we get the direction of the resultant?
- We can either find out Rx and R y.
- And then find out theta which is tan inverse of R y by R x where theta is the angle made by the resultant with the x axis.
- To find out the sense of the resultant, you have Rx and R Y, merge these two, whatever you are going to get will be the sense of the resultant.
- Then to determine the.
- Point of application of the resultant we should go for principle of movement or what we call it as the extension of variaus or law of movements.

- There could be a situation where we get only a moment as a resultant.
- We may not get the force as a resultant.
- I would take F1 and F2 and find out the resultant of these two, which would be this.
- Which would be this.
- That is, I just have completed the parallelogram.
- The diagon passing through the intersection of the two forces will give the resultant.
- This is the resultant R.
- But if this resultant that we get, okay, if that resultant of F1 and F2 that we are going to get as is will be equal to the magnitude of F3, which is acting as an external force on the rigid body.
- But in the opposite sense, as indicated here as minus F3, this would be the resultant of F1 and F2, which would be the same as the magnitude of F3, but the sense would be opposite.
- Then now you can see that these two.
- forms a couple and we know that the characteristics of a couple is the characteristics of a couple is to make the body rotate only and we also know that one of the.

Example or application from the lecture

- To find out what is the movement of all these three forces about this point O, which is given here.
- Okay now for that I have taken this example.
- Say for example we have F1, F2, and F3, three forces acting on a body.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000242 · EV0000243 · EV0000244 · +1 source units in pack.json

Core explanation · 15:00–19:59

characteristics of a couple S the net force acting because of a couple S zero.

Key details

- Like this, the other one is acting like this.
- The magnitude of that would be equal to F3 into D.
- F3 into D.
- So because now these two when I take it the moment produced will be a clockwise moment of magnitude which will be equal to F3 into D.
- And we also know about which axis it is acting, it is acting about the z axis which is coming out of the plane.
- No, where should we apply?
- We have already discussed that moment is a.
- Mom is a free vector.
- You can apply the moment anywhere you want.
- That would be the resultant of these three forces.
- Which is going to be acting.
- The resultant of these three forces now is just a moment of magnitude F3 into D, which would be acting in the clockwise direction along the z axis.

- You can apply it anywhere you want.
- The law of transmissibility of forces will apply, or the principle of transmissibility of forces will apply.
- The point of application of the vector can be anywhere, provided it is in contact with the body.
- Then only the body will experience that.
- I hope it is clear to everybody.
- In today's class, we have discussed the last topic of unit one, which is under chapter two.
- That is resulant.
- Resultant is that single force which produces the same effect as that of a system of forces.
- How do we find the resultant?
- The magnitude of the resultant is nothing but the dot product of all the forces.
- The direction is nothing but tan inverse of R y by Rx, provided the direction is with respect to the x axis.
- Where R is the component of all the forces along Y, and Rx which is the component of all the forces along X, R it is the component of the resultant about X and Y.
- Then we also have seen how to get the sense, you merge the Rx and R y together, you get the sense of the resultant R.
- Then to determine the point of application of the single force, you have to go for extension of Arianov's principle or principle of moments or law of moments, which states that the moment produced by a system of forces is equal to the moment produced by the.
- Resultant about the same point.
- And we have also looked at what could be the result.
- Resultant could be that only one single force or it could be a movement also.
- That what how do we get only one single movement is what I have explained in this last slide in this class.
- We will be continuing our class in the next class, that is class number 15, where I will take up and solve few problems related to resultant.
- We will meet in the next class.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000246 · EV0000247 · EV0000248 · +1 source units in pack.json

16 UE26ME142A_Unit 1_Class 12.0_AVSummary

Study focus: UE26ME142A_Unit 1_Class 12.0_AVSummary. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000250 · EV0000251

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

I am Dr.

Key details

- SV Venkatesh, Civil Engineering, Department of Science and Humanities, PS University.
- This is class number 15.
- In the last class, I had introduced you to the last topic of unit 1, chapter 2, that is resultant.
- We had discussed about what is resultant, how do we find the characteristics of resultant, that is the magnitude.
- The direction, the sense, and the point of application.
- The point of application can be determined by using the principle of movements.
- I had defined what is principle of moments and explained how do we find out the point of application of the resultant force for our system of forces.
- Solve a few problems taken from the unsolved problem category from the prescribed textbook that is Engineering Mechanics Statics by Miriam Craig and Bolton published by John Wiley Publications.
- The problem that I have chosen is 2 bar 81.
- The statement of the problem goes something like this.
- Replace the three horizontal forces and apply the couple with an equivalent force couple system at O.
- By specifying the resultant R and the couple M naught, next determine the equation for the line of action of the standalone resultant force R.
- There are three horizontal forces acting and there is a couple that is acting.
- We have to replace this system, that is this three.
- Forces plus one couple by an equivalent force couple system.
- What we had earlier discussed under the topic couple, force couple system at point O.
- Then we have a couple here.
- Then we have to determine.
- The line of action of the standalone resultant force.
- Line of action of the standalone resultant force is what is the line of action of the single force where we need to apply that.
- For any value of X and Y.
- Where this single resultant force needs to be acting, so that we can get the same effect as that of these three forces and the couple.
- Which is acting at this point is of magnitude 4 newton.
- The sense of this is towards the right.
- It is acting at a distance of 1 meter along x axis.
- Look at the axis.
- X axis is not the horizontal axis here, but this axis, x axis is inclined to the vertical at an angle of 70 degrees.
- This is acting from right to left.
- The magnitude of that is 6 kilonewtons.
- Then we have the another force of magnitude 10 kilonewtons acting on the x-axis at this point.
- which is equal to four meters still here.
- Whereas this six kilometers it act is acting along the x axis at a distance of two point five meters.

- This four kilometers is acting at a distance of one meter along the x axis.
- That is above x axis.
- Whereas this six kilometer is acting below x axis by a distance of 0.3 meters.

Example or application from the lecture

- Now if you read the diagram that is given, the first force.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000250 · EV0000251 · EV0000252 · +1 source units in pack.json

Core explanation · 05:00–09:59

This small distances also would play a very important role.

Key details

- First thing that we need to do is find out the dot product of all the forces.
- That would be the single force which is acting.
- I can straight away apply that at home.
- Or the force couple system when we are finding out at O would be the force would be equal to 8 kilonewtons acting from left to right along the horizontal.
- The next thing is determine the movement of all these four vectors at O.
- We have to resolve this four kilometers.
- Along x axis and y axis, similarly 6 kilonoton also along x axis and y axis, along this axis and along this axis, and this 10 kilonoton also along x axis of and y axis, of course the x-axis component of this 10 kiloton will not produce any movement.
- But these two both X both the component along X axis and Y axis would produce a moment at W.
- Then having this.
- Force couple system at what I will now apply that single resultant force which is 8 kilonoton at some point, okay, some point here, such that this force should produce the same effect or same moment, whatever we would have got because of this four.
- When does not go?
- Now R the single resultant force as I have told you 10 plus 4 minus 6 which is 8 kilonewtons and it is positive because it is active to the to from left to right along the horizontal.
- But if I resolve that along x axis and y axis, so I'll get this as 8 cos 20i which will be along x, then plus eight sin twenty j which will be along y axis.
- Now to find out moment about O, first we will take this moment which is acting in the anti-clockwise direction.
- Then we will take the component of this force along vertical.
- Which will be four cause of this angle twenty degrees.
- Sorry, sine of this angle which is twenty degrees.
- The advantage of resolving this force along X and Y would be simply that we can determine the perpendicular distance of those components and then we can multiply.
- Then we will take this component along y, which will be 6 sin 20, which will produce an clockwise movement, so it is negative.

- Into the distance at which this is acting is 2.5, then plus the component of this force.
- Which will be 10 sin 20, which will be acting along vertical, which will produce an anti-clockwise movement.
- Next the component of the 4 kilometer acting along x axis.
- Will be 4 cos 20 into this perpendicular distance will be 0.3.
- That would produce a clockwise moment, so it is negative.
- That would also produce a clockwise moment, so that is also minus 6 cos.
- 20 into 0.3.
- Of course point three is unable to be seen here because of my photo.
- So if you take out the total of this, you will get it as 22.1 kilonewton meters positive, which means that it is in the counterclockwise direction.

Example or application from the lecture

- So the free body diagram is this, as I told you, this is 20 degrees, this is also 20 degrees, because they have given this as 70 degrees.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000254 · EV0000255 · EV0000256 · +1 source units in pack.json

Core explanation · 10:00–14:59

If you take the resultant force, if I find out the cross product of the position vector of R.

Key details

- Cross the R vector that should be equal to the moment produced at O.
- Value of this that is R, whatever we have got 7.52i plus 2.74 J.
- I can get when I cross multiply this, I'll get this as 2.74 minus 7.52 y, which is equal to 22.1.
- This is about the z axis.
- So that means I have to apply.
- Sorry, x to be equal to minus 2.64.
- That means at apply the single resultant force to the left of O at this point.
- Then to find out what would be the y component, if I apply that single resultant force along y, what would be that?
- I hope it is clear for all of you.
- So we will move on to the next one.
- If you look at this, I will get an equation as y equals 0.364x minus 2.94.
- This is the equation of the line of action of the resultant force.
- If I want the y distance, I will substitute x equals 0, I get the y component.
- Or if I just re-assemble this equation.
- In as y equals in terms of x, I get this as equation.
- This is the line of action of the single resultant force that I need to apply.
- I hope it is clear.
- Coming to the next problem, I have considered 2 bar 91.
- Again, in this, we have three forces, and we are first to find out what is the force couple system at A, that is this point, and then we have to determine what is the X-intercept, that

means what should be the distance along the x-axis from A.
- At which I can apply the single resultant force such that it produces the same effect.
- So again in this problem, first we need to find out the force couple system at A for that resolve this force along horizontal and vertical.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000258 · EV0000259 · EV0000260 · +1 source units in pack.json

Core explanation · 15:00–19:59

Take all the component along X, so we get only one component that is 3.2.

Key details

- This would be 30 degrees.
- Sine 30, okay that would be the Rx value.
- Then 3.2 cos 30 would be the vertical component of 3.2 acting at C.
- Plus the component of this 4, which is 3.2 cos 30 minus 4.8.
- Then to find out the moment at this point.
- Take this force, multiply with this distance, so 10 into 1.2, which will produce a clockwise moment.
- Then you take this force, which would produce an anti-clockwise moment.
- Into we have to take this distance which would be equal to 0.6.
- This is 0.6, this is 30 degrees, so sin 30 would be the vertical distance, so I multiply that.
- Then 3.2 cos 30 into 1.8 plus this distance would be 0.6 cos 30 degrees plus 1.2 would be the distance.
- Counterclockwise movement, so I add that.
- Then I will take this force which will be clockwise, so minus 4.8 into this total distance would be 1.2 plus 1.2 cross 30 degrees plus 0.9 will be the movement acting at E.
- Plus 3.2 sin 30i plus cos 30 j.
- That is 1.6i minus 12.03 J.
- Then 4.8 into 1.2 plus 1.2 cost 30 plus 0.9 as I have explained.
- That would also be clockwise, so if we have taken it as positive.
- Then minus 3.2 cost 30 into 0.6 sine 30, that is the.
- This is the horizontal component, and this is the vertical distance.
- That's why this is positive, this is positive.
- Counterclockwise is taken as negative.
- I hope it is clear.
- This is R.
- X component is positive, y component is negative, so we get this as R.
- And the moment MA which is clockwise, we have got this.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000262 · EV0000263 · EV0000264 · +1 source units in pack.json

Core explanation · 20:00–24:59

If you would have got a negative sign here, then it would have indicated that instead of taking it to the right, we should have taken it to the left.

Key details

- So we will stop today's class here and continue in the next class.
- I'll be solving some more problems.
- My name is Dr.
- Sv Vengtesh from Civil Engineering, Department of Science and Humanities, PS University.
- We'll meet in the next class.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000266 · EV0000267 · EV0000268 · +1 source units in pack.json

17 UE26ME142A_Unit 1_Class 12.1_AVSummary

Study focus: UE26ME142A_Unit 1_Class 12.1_AVSummary. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000270 · EV0000271

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

My name is Dr.

Key details

- SV Venkitish from Civil Engineering under Department of Science and Humanities, PES University.
- This is class number sixteen.
- In the last class, that is class number fifteen, we had discussed.
- Two problems taken from the unsolved problems from the prescribed textbook that is Engineering Mechanics Statics by Miriam, Craig and Bolton.
- And it is the eighth edition.
- Today we will be continuing with.
- Some more numericals.
- And this is the last topic from unit one.
- For today's discussion, I have considered a problem on 2 bar 98.
- The statement of the problem goes something like this.
- Five forces are applied to the beam, trolley as shown, determine the coordinates of the point on the y-axis through which the standalone resultant R must pass.
- If F is equal to 5 kilonewton and theta is equal to 30 degrees.
- The Y intercept of the standalone resultant.
- What are we supposed to find is the y-intercept or the coordinate of the point on the y-axis at which the standalone resultant needs to be applied.

- This force the magnitude is three kilonewton again, it is acting along the x axis to the left.
- Then coming to this force, the magnitude of the force is 0.6, again it is acting along y, but it is in towards the right, so positive x axis, then this force is 0.8 kilometers, again it is acting along the x axis, again it is positive because it is acting to the right.
- The angle at which the F force is acting with respect to the y-axis.
- Axis, you are supposed to take that as the positive axis.
- X axis positive is to the right and downwards is the positive y axis.
- Of course, your x-axis is the center of this, which is at 140 mm.
- how do we go about this?
- This is a free body diagram.
- Then the F force is acting here.
- Then of course this also is considered, which you are unable to see here.
- This is the free body diagram here.
- This distance is 280 mm.
- First take all the forces along acting along x axis, algebraic sum of all the forces.

Example or application from the lecture

- The force F magnitude is given as 5 kilo Newton and the angle T at which this force is acting is given as 30 degrees.
- F force, the magnitude of F is given as 5 kilometer.
- Is given as 30 degrees.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000270 · EV0000271 · EV0000272 · +1 source units in pack.json

Core explanation · 05:00–09:59

Then take all the forces acting along y, algebraic sum of that, then you get the component of the resultant about the y-axis.

Key details

- With the values that we have here, you get the component of the resultant force acting towards the left and along the x-axis and component of the force.
- Along y-axis will be acting downwards.
- You can also find out the angle at which the resultant is acting with respect to the negative x axis or the positive y axis.
- Once you get that.
- The next thing is we have to find out where are we supposed to apply this force along the y-axis for which we will go for usage of principle of movements or extension of Baring North principle, that is.
- Moment produced by all the forces about one point, where in this problem we will take it as a point O should be equal to the moment produced by the resultant force.
- If I take the resultant to be acting here, I will get the X component of the resultant into the distance at which it is acting.
- If I take it as y, then I can find out y.

- Then we have one force acting towards the right, that is point six, another force acting towards the right again, that is point eight.
- Then we also have the component of the.
- F force that is acting along X axis, which is B 5, sine 30 degrees, that will be acting to the right, so that is positive.
- Next, to take all the forces acting along y, f p will be.
- The component of the F4 which is acting along the y axis which will be downward that is positive, which will be phi cos 30 degrees, you will get that as 4.33 kilonutons.
- Now take the moment about the point O because of all the forces.
- You take the force 4 kilometer that is acting to the left side along the x-axis, the distance would be 280 divided by 2, which was the distance which I had talked about, plus we should take the radius of the pulley that we have or the diameter of the pulley at the top as well as at the bottom.
- Then plus 3 kilonutan is acting at the bottom of the pulley, top pulley.
- Plus we have 0.6, which is also going to be acting at the downward, which is at a distance of 280 by 2 itself.
- This would also produce an anti-clockwise movement.
- Plus 0.8 would also produce an anti-clock point moment, but the distance would be 280 by 2 plus 55 plus 55.
- Because I have taken this in the millimeters on this force magnitude in kilone, so it is kilonewton millimeters.
- Applied this resultant force.
- On the x-axis, okay.
- To the left of O, then you will have a negative movement, so you get that to be equal to 393.53 mm.
- Or I'll apply this resultant force, this resultant force on the y-axis, which is below O.

Example or application from the lecture

- figure that is given you can make out what is the radius of the pulley.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000274 · EV0000275 · EV0000276 · +1 source units in pack.json

Core explanation · 10:00–14:59

Minus this would be minus 3.1, which will be acting to the left.

Key details

- That would be the.
- Y intercept of the resultant from the point O.
- Next, I have taken the last problem in this unit, unit 1, chapter 2.
- That is problem 2 bar 100.
- The problem statement goes something like this.
- An exhaust system for a pickup truck is shown in the figure.
- The weights WH, WM under W1 of the head pipe.

- Moffl and tailpipe are 10, 100, comma 50 respectively.
- And at the indicated points, if the exhaust pipe hanger.
- At point A is adjusted so that its tension FA is 50 newtons.
- Determine the required forces in the hangar at point B, C and D.
- Why is a zero force couple system at O is desirable.
- The weight of WH, which is this weight, the weight of W M, which is this one, the weight of W1, which is this one.
- And they have asked you to determine the force at B, that is this force F B, F C, and F such that the force couple system at this point O is equal to 0.
- If the exhaust pipe hanger at point A is adjusted so that its tension FA is 50 newton, why is the zero force couple system what most desirable?
- That is what is asked.
- This would be a candile.
- If the forces are balanced with each other, there won't be any forces that is acting at war.
- That is the first thing why a force couple system, zero force couple is desirable.
- And there is no force acting at this point.
- And it should also not rotate.
- That is why your couple at this point should also be zero.
- That is the explanation for why we should not.
- Have or have or why we should have zero force couple system.
- And also we can note down that the force acting at AB, then your WH, WM, and WT, all of them are acting vertically.
- Except that F C and F D are acting.
- At an angle to the vertical, which is 30 degrees for both of them.
- That means the component of the resultant force along X and along Y should be zero.

Example or application from the lecture

- Or it is W T it should be, I think I'm sorry, this is W T, so this should be W T are given respectively as 10, 100 and 50.
- They have also given the force acting at A, that is this force F A to be equal to 50 newton.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000278 · EV0000279 · EV0000280 · +1 source units in pack.json

Core explanation · 15:00–19:59

X axis force, whereas Fd will produce a positive x axis force.

Key details

- Next, if I take all the forces acting along the vertical.
- I can say that this component F C and F D plus this component F C minus these three forces would give me the force in F B.
- Then to determine the value of F C and F D, I will take the moment about O, which we will get it in terms of F P, F C and F D.
- One is force at B, which we have to find, force C we have to find, force at B we have to find.

- There are three unknowns, so at least we require three equations.
- Which are these three equations?
- Since the resultant at O is 0, the component of the resultant along X should be equal to 0.
- The rest is acting along the vertical.
- Minus 10 is your one force that is W acting.
- Then one more force.
- W this is WH this is W M and this is W T all the three of them are acting downwards so it is negative Your F B is acting upwards.
- The component of the force F C along the vertical will be acting upwards, so positive, that is F C cos 30 degrees.
- The component of the force F T will also be acting upwards, that is F T cos 30 degrees acting upwards.
- From this, if I take all the components acting along the y, I can easily make out that F C is equal to F D, which is equal to F.
- If I take all the component acting along the y direction, substituting this here for F C and F D as F, I can get the value of F P in terms of F.
- To all the unknowns to be acting or in terms of F, I can make out.
- Next you take movement about O of all the forces.
- 50 is the force acting at A, which is acting upwards.
- 100 is M B which is acting downward, which will produce a clockwise movement, the distance from O to B or where the force is acting F.
- f you will have the distance as 1.35 then your f p force is acting upward which will produce an anti clockwise movement the distance from o to that force is two kilon two meters so that is what is taken here then your f.

Example or application from the lecture

- As per the statement given, we are supposed to find out the force at B, force at C, and force at D.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000282 · EV0000283 · EV0000284 · +1 source units in pack.json

Core explanation · 20:00–24:59

T is acting downward, which will produce a clockwise movement.

Key details

- Then we have the two forces F C and F D, the component of those two, you can take it as 2F, cos 30 degrees acting vertically, the distance of that would be 2.9.
- Here I can substitute this in this equation, and then I can get the value of F B.
- Then substitute that whatever relationship that we have got, F B in terms of F, then substitute.
- Here we get the value of F C and F D.
- I hope this is also clear.
- In the next class, we will have an experiential component class, that is class number 17.

- Next to that, we will start with the next unit, that is unit 2.
- That unit is equilibrium chapter.
- I am Dr.
- SV Ventesh from Civil Engineering under Department of Science and Humanities, PS University.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000286 · EV0000287 · EV0000288 · +1 source units in pack.json

18 Source slide deck: UE26ME142A Unit 1 Class 1.0 Lecture PPT

Study focus: Source slide deck: UE26ME142A Unit 1 Class 1.0 Lecture PPT. The notes identify each page heading, summarize its main ideas, and explain its place in this unit; citations preserve the source page locator. EV0000290 · EV0000291

SHORT-NOTE RECAP

Study notes · Source slide deck: UE26ME142A Unit 1 Class 1.0 Lecture PPT

The slide deck is rewritten here as searchable lesson notes. Each subsection states the slide heading and its concise teaching points; page citations lead back to the indexed source image when visual details are needed.

Slide 1 · Slide page 1

- no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection.

Slide 2 · Engineering Mechanics

- J. L. Meriam and L. G. Kraige
- Copyright © 2012 by John Wiley & Sons, Inc. All rights
- S V Venkatesh - Engineering Mechanics Statics - Unit 1 - Introduction

Slide 3 · ENGINEERING MECHANICS STATICS

- S V Venkatesh - Engineering Mechanics Statics - Unit 1 - Introduction

Slide 4 · ENGINEERING MECHANICS STATICS

- S V Venkatesh - Engineering Mechanics Statics - Unit 1 - Introduction

Slide 5 · Slide page 5

- no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection.

EV0000290 · EV0000291 · EV0000292 · +8 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Official unit catalogue	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/units.json	Official course/unit title and index.

Source	Path	Use
UE26ME142A_Unit 1_Class 1.1_AVSummary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/001_UE26ME142A_Unit_1_Class_1.1_AVSummary__bea4975b08967ab4.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 1.1_AVSummary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/001_UE26ME142A_Unit_1_Class_1.1_AVSummary__bea4975b08967ab4.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 1.1_AVSummary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/001_UE26ME142A_Unit_1_Class_1.1_AVSummary__bea4975b08967ab4.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9993; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 1.1_AVSummary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/001_UE26ME142A_Unit_1_Class_1.1_AVSummary__bea4975b08967ab4.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 1_AVSummary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/001_UE26ME142A_Unit_1_Class_1_AVSummary__1ad410489823544f.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 1_AVSummary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/001_UE26ME142A_Unit_1_Class_1_AVSummary__1ad410489823544f.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
UE26ME142A_Unit 1_Class 1_AVSummary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/001_UE26ME142A_Unit_1_Class_1_AVSummary__1ad410489823544f.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9994; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 1_AVSummary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/001_UE26ME142A_Unit_1_Class_1_AVSummary__1ad410489823544f.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 2.0_AVSummary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/002_UE26ME142A_Unit_1_Class_2.0_AVSummary__254d366ff7b73c45.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 2.0_AVSummary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/002_UE26ME142A_Unit_1_Class_2.0_AVSummary__254d366ff7b73c45.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 2.0_AVSummary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/002_UE26ME142A_Unit_1_Class_2.0_AVSummary__254d366ff7b73c45.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9994; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 2.0_AVSummary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/002_UE26ME142A_Unit_1_Class_2.0_AVSummary__254d366ff7b73c45.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
UE26ME142A_Unit 1_Class 2.1_AVSummary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/002_UE26ME142A_Unit_1_Class_2.1_AVSummary__3c66109b736f8c89.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 2.1_AVSummary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/002_UE26ME142A_Unit_1_Class_2.1_AVSummary__3c66109b736f8c89.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 2.1_AVSummary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/002_UE26ME142A_Unit_1_Class_2.1_AVSummary__3c66109b736f8c89.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9993; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 2.1_AVSummary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/002_UE26ME142A_Unit_1_Class_2.1_AVSummary__3c66109b736f8c89.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 2.2_AVSummary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/002_UE26ME142A_Unit_1_Class_2.2_AVSummary__80ec6ef9f2cc872e.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 2.2_AVSummary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/002_UE26ME142A_Unit_1_Class_2.2_AVSummary__80ec6ef9f2cc872e.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
UE26ME142A_Unit 1_Class 2.2_AVSummary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/002_UE26ME142A_Unit_1_Class_2.2_AVSummary__80ec6ef9f2cc872e.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9993; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 2.2_AVSummary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/002_UE26ME142A_Unit_1_Class_2.2_AVSummary__80ec6ef9f2cc872e.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 3_AVSummary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/003_UE26ME142A_Unit_1_Class_3_AVSummary__eee232e254abeb39.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 3_AVSummary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/003_UE26ME142A_Unit_1_Class_3_AVSummary__eee232e254abeb39.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 3_AVSummary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/003_UE26ME142A_Unit_1_Class_3_AVSummary__eee232e254abeb39.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9996; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 3_AVSummary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/003_UE26ME142A_Unit_1_Class_3_AVSummary__eee232e254abeb39.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
UE26ME142A_Unit 1_Class 4_AVSummary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/004_UE26ME142A_Unit_1_Class_4_AVSummary__93feea637b67e79c.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 4_AVSummary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/004_UE26ME142A_Unit_1_Class_4_AVSummary__93feea637b67e79c.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 4_AVSummary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/004_UE26ME142A_Unit_1_Class_4_AVSummary__93feea637b67e79c.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9995; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 4_AVSummary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/004_UE26ME142A_Unit_1_Class_4_AVSummary__93feea637b67e79c.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 5_AVSummary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/005_UE26ME142A_Unit_1_Class_5_AVSummary__86fffcdbe06dbb6b.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 5_AVSummary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/005_UE26ME142A_Unit_1_Class_5_AVSummary__86fffcdbe06dbb6b.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
UE26ME142A_Unit 1_Class 5_AVSummary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/005_UE26ME142A_Unit_1_Class_5_AVSummary__86fffcdbe06dbb6b.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9996; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 5_AVSummary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/005_UE26ME142A_Unit_1_Class_5_AVSummary__86fffcdbe06dbb6b.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 6_AVSummary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/006_UE26ME142A_Unit_1_Class_6_AVSummary__f35d743d3b761113.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 6_AVSummary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/006_UE26ME142A_Unit_1_Class_6_AVSummary__f35d743d3b761113.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 6_AVSummary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/006_UE26ME142A_Unit_1_Class_6_AVSummary__f35d743d3b761113.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9995; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 6_AVSummary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/006_UE26ME142A_Unit_1_Class_6_AVSummary__f35d743d3b761113.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
UE26ME142A_Unit 1_Class 8_AVSummary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/008_UE26ME142A_Unit_1_Class_8_AVSummary__f49bcc7a576db09d.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 8_AVSummary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/008_UE26ME142A_Unit_1_Class_8_AVSummary__f49bcc7a576db09d.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 8_AVSummary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/008_UE26ME142A_Unit_1_Class_8_AVSummary__f49bcc7a576db09d.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9995; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 8_AVSummary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/008_UE26ME142A_Unit_1_Class_8_AVSummary__f49bcc7a576db09d.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 9_AVSummary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/009_UE26ME142A_Unit_1_Class_9_AVSummary__866a19feaac2a3bf.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 9_AVSummary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/009_UE26ME142A_Unit_1_Class_9_AVSummary__866a19feaac2a3bf.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
UE26ME142A_Unit 1_Class 9_AVSummary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/009_UE26ME142A_Unit_1_Class_9_AVSummary__866a19feaac2a3bf.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9994; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 9_AVSummary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/009_UE26ME142A_Unit_1_Class_9_AVSummary__866a19feaac2a3bf.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 10.0_AVSummary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/010_UE26ME142A_Unit_1_Class_10.0_AVSummary__4d79e3bc52cb3c4f.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 10.0_AVSummary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/010_UE26ME142A_Unit_1_Class_10.0_AVSummary__4d79e3bc52cb3c4f.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 10.0_AVSummary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/010_UE26ME142A_Unit_1_Class_10.0_AVSummary__4d79e3bc52cb3c4f.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9993; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 10.0_AVSummary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/010_UE26ME142A_Unit_1_Class_10.0_AVSummary__4d79e3bc52cb3c4f.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
UE26ME142A_Unit 1_Class 10.1_AVSummary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/010_UE26ME142A_Unit_1_Class_10.1_AVSummary__c39186c34db5b2f5.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 10.1_AVSummary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/010_UE26ME142A_Unit_1_Class_10.1_AVSummary__c39186c34db5b2f5.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 10.1_AVSummary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/010_UE26ME142A_Unit_1_Class_10.1_AVSummary__c39186c34db5b2f5.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9993; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 10.1_AVSummary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/010_UE26ME142A_Unit_1_Class_10.1_AVSummary__c39186c34db5b2f5.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 11_AVSummary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/011_UE26ME142A_Unit_1_Class_11_AVSummary__3d8f98864b87c9b2.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 11_AVSummary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/011_UE26ME142A_Unit_1_Class_11_AVSummary__3d8f98864b87c9b2.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
UE26ME142A_Unit 1_Class 11_AVSummary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/011_UE26ME142A_Unit_1_Class_11_AVSummary__3d8f98864b87c9b2.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9995; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 11_AVSummary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/011_UE26ME142A_Unit_1_Class_11_AVSummary__3d8f98864b87c9b2.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 12.0_AVSummary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/012_UE26ME142A_Unit_1_Class_12.0_AVSummary__ff48ce3f9a237d73.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 12.0_AVSummary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/012_UE26ME142A_Unit_1_Class_12.0_AVSummary__ff48ce3f9a237d73.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 12.0_AVSummary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/012_UE26ME142A_Unit_1_Class_12.0_AVSummary__ff48ce3f9a237d73.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9993; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 12.0_AVSummary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/012_UE26ME142A_Unit_1_Class_12.0_AVSummary__ff48ce3f9a237d73.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
UE26ME142A_Unit 1_Class 12.1_AVSummary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/012_UE26ME142A_Unit_1_Class_12.1_AVSummary__4e48a92aba0ceae0.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 12.1_AVSummary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/012_UE26ME142A_Unit_1_Class_12.1_AVSummary__4e48a92aba0ceae0.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 12.1_AVSummary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/012_UE26ME142A_Unit_1_Class_12.1_AVSummary__4e48a92aba0ceae0.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9994; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 12.1_AVSummary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/012_UE26ME142A_Unit_1_Class_12.1_AVSummary__4e48a92aba0ceae0.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Slide page text: UE26ME142A Unit 1 Class 1.0 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/001_UE26ME142A_Unit_1_Class_1.0_Lecture_PPT__a53eb8b08a5ac7ff.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE26ME142A Unit 1 Class 1.0 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/001_UE26ME142A_Unit_1_Class_1.0_Lecture_PPT__a53eb8b08a5ac7ff.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

Source	Path	Use
Page-render source for UE26ME142A Unit 1 Class 1.0 Lecture PPT (converted PDF; original hash 1adeec92bd8b55d28a301ab05f3ac6211855294e88abc2dc8c745e99c5a081a1)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/001_UE26ME142A_Unit_1_Class_1.0_Lecture_PPT__a53eb8b08a5ac7ff.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE26ME142A Unit 1 Class 1.1 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/001_UE26ME142A_Unit_1_Class_1.1_Lecture_PPT__a6f76a088910b54a.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE26ME142A Unit 1 Class 1.1 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/001_UE26ME142A_Unit_1_Class_1.1_Lecture_PPT__a6f76a088910b54a.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for UE26ME142A Unit 1 Class 1.1 Lecture PPT (converted PDF; original hash 1653a8ffbc63c6f1cebca6688bc6f0c7cee6632d7af40bab1d99597d8dbeaedb)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/001_UE26ME142A_Unit_1_Class_1.1_Lecture_PPT__a6f76a088910b54a.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE26ME142A Unit 1 Class 2.0 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/002_UE26ME142A_Unit_1_Class_2.0_Lecture_PPT__de74ea957f21118c.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE26ME142A Unit 1 Class 2.0 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/002_UE26ME142A_Unit_1_Class_2.0_Lecture_PPT__de74ea957f21118c.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

Source	Path	Use
Page-render source for UE26ME142A Unit 1 Class 2.0 Lecture PPT (converted PDF; original hash 3661bb44d6ba64ef223560e46dbb47020bbbbd8adba40bdebe6abc98c692f8ea)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/002_UE26ME142A_Unit_1_Class_2.0_Lecture_PPT__de74ea957f21118c.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE26ME142A Unit 1 Class 2.1 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/002_UE26ME142A_Unit_1_Class_2.1_Lecture_PPT__bb478549c01cfb60.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE26ME142A Unit 1 Class 2.1 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/002_UE26ME142A_Unit_1_Class_2.1_Lecture_PPT__bb478549c01cfb60.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for UE26ME142A Unit 1 Class 2.1 Lecture PPT (converted PDF; original hash 36e0e092882d543954d5d699ebdfc177ff83bb355782a7a41956ee4b6874289d)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/002_UE26ME142A_Unit_1_Class_2.1_Lecture_PPT__bb478549c01cfb60.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE26ME142A Unit 1 Class 2.2 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/002_UE26ME142A_Unit_1_Class_2.2_Lecture_PPT__3fc8761d12196c6f.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE26ME142A Unit 1 Class 2.2 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/002_UE26ME142A_Unit_1_Class_2.2_Lecture_PPT__3fc8761d12196c6f.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

Source	Path	Use
Page-render source for UE26ME142A Unit 1 Class 2.2 Lecture PPT (converted PDF; original hash a6f22c4d1cac09e09f87cfafbf1e99a0ddc24850719cb5b5ce4ed918d7d5ab9e)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/002_UE26ME142A_Unit_1_Class_2.2_Lecture_PPT__3fc8761d12196c6f.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE26ME142A Unit 1 Class 3 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/003_UE26ME142A_Unit_1_Class_3_Lecture_PPT__40166a1710a03d41.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE26ME142A Unit 1 Class 3 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/003_UE26ME142A_Unit_1_Class_3_Lecture_PPT__40166a1710a03d41.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for UE26ME142A Unit 1 Class 3 Lecture PPT (converted PDF; original hash b0f98082451c388673f5a78dde1aea22a173f2b5d16752ebf90f8207e81c713d)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/003_UE26ME142A_Unit_1_Class_3_Lecture_PPT__40166a1710a03d41.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE26ME142A Unit 1 Class 4 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/004_UE26ME142A_Unit_1_Class_4_Lecture_PPT__984b7c368532af06.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE26ME142A Unit 1 Class 4 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/004_UE26ME142A_Unit_1_Class_4_Lecture_PPT__984b7c368532af06.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

Source	Path	Use
Page-render source for UE26ME142A Unit 1 Class 4 Lecture PPT (converted PDF; original hash e14ad2ad44deec3d7e8473a5dcb210d1d4719a1ab45ba54c227fc13ef666ddb7)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/004_UE26ME142A_Unit_1_Class_4_Lecture_PPT__984b7c368532af06.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE26ME142A Unit 1 Class 5 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/005_UE26ME142A_Unit_1_Class_5_Lecture_PPT__f49d361d79853022.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE26ME142A Unit 1 Class 5 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/005_UE26ME142A_Unit_1_Class_5_Lecture_PPT__f49d361d79853022.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for UE26ME142A Unit 1 Class 5 Lecture PPT (converted PDF; original hash 3b5cb4980cdc4b41b4f2e156137ff96a23026d13ab15abde89149d8d720351d9)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/005_UE26ME142A_Unit_1_Class_5_Lecture_PPT__f49d361d79853022.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE26ME142A Unit 1 Class 6 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/006_UE26ME142A_Unit_1_Class_6_Lecture_PPT__f79618b4a0f260c2.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE26ME142A Unit 1 Class 6 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/006_UE26ME142A_Unit_1_Class_6_Lecture_PPT__f79618b4a0f260c2.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

Source	Path	Use
Page-render source for UE26ME142A Unit 1 Class 6 Lecture PPT (converted PDF; original hash 22dd11615d21af38626b1a33b7b1d8446e3a14b0733b41776fbeab98198f7285)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/006_UE26ME142A_Unit_1_Class_6_Lecture_PPT__f79618b4a0f260c2.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE26ME142A Unit 1 Class 8 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/008_UE26ME142A_Unit_1_Class_8_Lecture_PPT__2659a840cc2bfd13.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE26ME142A Unit 1 Class 8 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/008_UE26ME142A_Unit_1_Class_8_Lecture_PPT__2659a840cc2bfd13.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for UE26ME142A Unit 1 Class 8 Lecture PPT (converted PDF; original hash 1c1d191c0ed477478825d748b36f9d8c40417a73e583b877f3773c82d759991c)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/008_UE26ME142A_Unit_1_Class_8_Lecture_PPT__2659a840cc2bfd13.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE26ME142A Unit 1 Class 9 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/009_UE26ME142A_Unit_1_Class_9_Lecture_PPT__832f697eaf1d383c.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE26ME142A Unit 1 Class 9 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/009_UE26ME142A_Unit_1_Class_9_Lecture_PPT__832f697eaf1d383c.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

Source	Path	Use
Page-render source for UE26ME142A Unit 1 Class 9 Lecture PPT (converted PDF; original hash 568f6e5391b7defd2b7d1afe3863f7da14ce323d0de3008413f995d3c130cdb4)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/009_UE26ME142A_Unit_1_Class_9_Lecture_PPT__832f697eaf1d383c.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE26ME142A Unit 1 Class 10.0 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/010_UE26ME142A_Unit_1_Class_10.0_Lecture_PPT__aa8f331eb00d810f.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE26ME142A Unit 1 Class 10.0 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/010_UE26ME142A_Unit_1_Class_10.0_Lecture_PPT__aa8f331eb00d810f.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for UE26ME142A Unit 1 Class 10.0 Lecture PPT (converted PDF; original hash f9dc99f62b1b462ef1ce3968961567 17fad583b69f6eefddb607e7fcf04a1c15)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/010_UE26ME142A_Unit_1_Class_10.0_Lecture_PPT__aa8f331eb00d810f.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE26ME142A Unit 1 Class 10.1 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/010_UE26ME142A_Unit_1_Class_10.1_Lecture_PPT__fe73ddc3ca596069.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE26ME142A Unit 1 Class 10.1 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/010_UE26ME142A_Unit_1_Class_10.1_Lecture_PPT__fe73ddc3ca596069.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

Source	Path	Use
Page-render source for UE26ME142A Unit 1 Class 10.1 Lecture PPT (converted PDF; original hash 3dd0700564b31ad9ccb79167f4be5e9b2d3f520ab58af273220b2f5400baf9cb)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/010_UE26ME142A_Unit_1_Class_10.1_Lecture_PPT__fe73ddc3ca596069.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE26ME142A Unit 1 Class 11 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/011_UE26ME142A_Unit_1_Class_11_Lecture_PPT__3624142116c7273d.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE26ME142A Unit 1 Class 11 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/011_UE26ME142A_Unit_1_Class_11_Lecture_PPT__3624142116c7273d.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for UE26ME142A Unit 1 Class 11 Lecture PPT (converted PDF; original hash 8de1359871fb43079aea2ca316a794b5f1f0ff2e8b79bf892009259a7d41b25d)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/011_UE26ME142A_Unit_1_Class_11_Lecture_PPT__3624142116c7273d.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE26ME142A Unit 1 Class 12.0 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/012_UE26ME142A_Unit_1_Class_12.0_Lecture_PPT__2e7b61b526d51f60.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE26ME142A Unit 1 Class 12.0 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/012_UE26ME142A_Unit_1_Class_12.0_Lecture_PPT__2e7b61b526d51f60.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

Source	Path	Use
Page-render source for UE26ME142A Unit 1 Class 12.0 Lecture PPT (converted PDF; original hash f358c92b19e16b9e23427a6f5aa33c6a2d5811e70282189b6a221316ec336963)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/012_UE26ME142A_Unit_1_Class_12.0_Lecture_PPT__2e7b61b526d51f60.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE26ME142A Unit 1 Class 12.1 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/012_UE26ME142A_Unit_1_Class_12.1_Lecture_PPT__1d60317fe4af0cbe.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE26ME142A Unit 1 Class 12.1 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/012_UE26ME142A_Unit_1_Class_12.1_Lecture_PPT__1d60317fe4af0cbe.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for UE26ME142A Unit 1 Class 12.1 Lecture PPT (converted PDF; original hash ec9d7e8c8c05cfbfda48f6154f40ac51ad134a3db1ca4c0c31537c2fe1f2670a)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/012_UE26ME142A_Unit_1_Class_12.1_Lecture_PPT__1d60317fe4af0cbe.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 2 rendered pages in 001_UE26ME142A_Unit_1_Class_1.0_Lecture_PPT__a53eb8b08a5ac7ff.slides.md have no readable `.slides.md` text; their original page images are included.
2. 2 rendered pages in 001_UE26ME142A_Unit_1_Class_1.1_Lecture_PPT__a6f76a088910b54a.slides.md have no readable `.slides.md` text; their original page images are included.
3. 2 rendered pages in 002_UE26ME142A_Unit_1_Class_2.0_Lecture_PPT__de74ea957f21118c.slides.md have no readable `.slides.md` text; their original page images are included.

4. 2 rendered pages in 002_UE26ME142A_Unit_1_Class_2.1_Lecture_PPT__bb478549c01cfb60.slides.md have no readable `.slides.md` text; their original page images are included.
5. 2 rendered pages in 002_UE26ME142A_Unit_1_Class_2.2_Lecture_PPT__3fc8761d12196c6f.slides.md have no readable `.slides.md` text; their original page images are included.
6. 3 rendered pages in 003_UE26ME142A_Unit_1_Class_3_Lecture_PPT__40166a1710a03d41.slides.md have no readable `.slides.md` text; their original page images are included.
7. 3 rendered pages in 004_UE26ME142A_Unit_1_Class_4_Lecture_PPT__984b7c368532af06.slides.md have no readable `.slides.md` text; their original page images are included.
8. 3 rendered pages in 005_UE26ME142A_Unit_1_Class_5_Lecture_PPT__f49d361d79853022.slides.md have no readable `.slides.md` text; their original page images are included.
9. 2 rendered pages in 006_UE26ME142A_Unit_1_Class_6_Lecture_PPT__f79618b4a0f260c2.slides.md have no readable `.slides.md` text; their original page images are included.
10. 3 rendered pages in 008_UE26ME142A_Unit_1_Class_8_Lecture_PPT__2659a840cc2bfd13.slides.md have no readable `.slides.md` text; their original page images are included.
11. 2 rendered pages in 009_UE26ME142A_Unit_1_Class_9_Lecture_PPT__832f697eaf1d383c.slides.md have no readable `.slides.md` text; their original page images are included.
12. 2 rendered pages in 010_UE26ME142A_Unit_1_Class_10.0_Lecture_PPT__aa8f331eb00d810f.slides.md have no readable `.slides.md` text; their original page images are included.
13. 2 rendered pages in 010_UE26ME142A_Unit_1_Class_10.1_Lecture_PPT__fe73ddc3ca596069.slides.md have no readable `.slides.md` text; their original page images are included.
14. 2 rendered pages in 011_UE26ME142A_Unit_1_Class_11_Lecture_PPT__3624142116c7273d.slides.md have no readable `.slides.md` text; their original page images are included.
15. 2 rendered pages in 012_UE26ME142A_Unit_1_Class_12.0_Lecture_PPT__2e7b61b526d51f60.slides.md have no readable `.slides.md` text; their original page images are included.
16. 4 rendered pages in 012_UE26ME142A_Unit_1_Class_12.1_Lecture_PPT__1d60317fe4af0cbe.slides.md have no readable `.slides.md` text; their original page images are included.
17. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.

18. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
19. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
20. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
21. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00001	1 covered · 0 excluded · 0 gaps	catalogue: Unit title and index copied from the official units.json catalogue; human content review is pending.
SRC00002	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0059. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00003	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0059. Sidecar variants compare...
SRC00004	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0059. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00005	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0059. Sidecar variants compare...
SRC00006	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0036. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0037 to s0052. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00007	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0036. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0037 to s0052. Sidecar variants compare...
SRC00008	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0036. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0037 to s0052. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00009	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0036. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0037 to s0052. Sidecar variants compare...
SRC00010	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0062. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00011	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0062. Sidecar variants compare...
SRC00012	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0062. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00013	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0062. Sidecar variants compare...
SRC00014	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0058. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00015	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0058. Sidecar variants compare...
SRC00016	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0058. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00017	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0058. Sidecar variants compare...
SRC00018	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0057. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00019	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0057. Sidecar variants compare...
SRC00020	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0057. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00021	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0057. Sidecar variants compare...
SRC00022	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00023	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare...
SRC00024	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00025	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare...
SRC00026	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0060. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00027	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0060. Sidecar variants compare...
SRC00028	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0060. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00029	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0060. Sidecar variants compare...
SRC00030	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0056. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00031	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0056. Sidecar variants compare...
SRC00032	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0056. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00033	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0056. Sidecar variants compare...
SRC00034	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0053. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00035	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0053. Sidecar variants compare...
SRC00036	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0053. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00037	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0053. Sidecar variants compare...
SRC00038	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0057. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00039	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0057. Sidecar variants compare...
SRC00040	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0057. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00041	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0057. Sidecar variants compare...
SRC00042	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0055. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00043	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0055. Sidecar variants compare...
SRC00044	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0055. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00045	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0055. Sidecar variants compare...
SRC00046	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0060. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00047	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0060. Sidecar variants compare...
SRC00048	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0060. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00049	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0060. Sidecar variants compare...
SRC00050	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0054. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00051	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0054. Sidecar variants compare...
SRC00052	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0054. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00053	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0054. Sidecar variants compare...
SRC00054	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0036. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0037 to s0055. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00055	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0036. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0037 to s0055. Sidecar variants compare...
SRC00056	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0036. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0037 to s0055. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00057	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0036. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0037 to s0055. Sidecar variants compare...
SRC00058	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0036. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0037 to s0053. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00059	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0036. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0037 to s0053. Sidecar variants compare...
SRC00060	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0036. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0037 to s0053. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00061	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0036. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0037 to s0053. Sidecar variants compare...
SRC00062	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0057. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00063	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0057. Sidecar variants compare...
SRC00064	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0057. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00065	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0057. Sidecar variants compare...
SRC00066	3 covered · 0 excluded · 0 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00068	5 covered · 0 excluded · 0 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00067	3 covered · 0 excluded · 0 gaps	p1: Alternate extracted slide text page 1; text differs from or is absent in the .slides.md page;... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...

Source	Unit counts	Example decisions
SRC00069	0 covered · 0 excluded · 2 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from...
SRC00071	0 covered · 0 excluded · 4 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00070	0 covered · 0 excluded · 2 gaps	p1: Alternate extracted slide text page 1; text differs from or is absent in the .slides.md page;... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human...
SRC00072	0 covered · 0 excluded · 3 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00074	0 covered · 0 excluded · 5 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...

Source	Unit counts	Example decisions
SRC00073	0 covered · 0 excluded · 3 gaps	p1: Alternate extracted slide text page 1; text differs from or is absent in the .slides.md page;... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00075	0 covered · 0 excluded · 9 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00077	0 covered · 0 excluded · 11 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00076	0 covered · 0 excluded · 9 gaps	p1: Alternate extracted slide text page 1; text differs from or is absent in the .slides.md page;... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00078	0 covered · 0 excluded · 11 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...

Source	Unit counts	Example decisions
SRC00080	0 covered · 0 excluded · 13 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00079	0 covered · 0 excluded · 11 gaps	p1: Alternate extracted slide text page 1; text differs from or is absent in the .slides.md page;... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00081	0 covered · 0 excluded · 9 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00083	0 covered · 0 excluded · 12 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...

Source	Unit counts	Example decisions
SRC00082	0 covered · 0 excluded · 9 gaps	p1: Alternate extracted slide text page 1; text differs from or is absent in the .slides.md page;... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00084	0 covered · 0 excluded · 3 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00086	0 covered · 0 excluded · 6 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00085	0 covered · 0 excluded · 3 gaps	p1: Alternate extracted slide text page 1; text differs from or is absent in the .slides.md page;... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; text differs from or is absent in the .slides.md page;...
SRC00087	0 covered · 0 excluded · 6 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...

Source	Unit counts	Example decisions
SRC00089	0 covered · 0 excluded · 9 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00088	0 covered · 0 excluded · 6 gaps	p1: Alternate extracted slide text page 1; text differs from or is absent in the .slides.md page;... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00090	0 covered · 0 excluded · 7 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00092	0 covered · 0 excluded · 9 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...

Source	Unit counts	Example decisions
SRC00091	0 covered · 0 excluded · 7 gaps	p1: Alternate extracted slide text page 1; text differs from or is absent in the .slides.md page;... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00093	0 covered · 0 excluded · 8 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00095	0 covered · 0 excluded · 11 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00094	0 covered · 0 excluded · 8 gaps	p1: Alternate extracted slide text page 1; text differs from or is absent in the .slides.md page;... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; text differs from or is absent in the .slides.md page;...
SRC00096	0 covered · 0 excluded · 6 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...

Source	Unit counts	Example decisions
SRC00098	0 covered · 0 excluded · 8 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00097	0 covered · 0 excluded · 6 gaps	p1: Alternate extracted slide text page 1; text differs from or is absent in the .slides.md page;... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00099	0 covered · 0 excluded · 2 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from...
SRC00101	0 covered · 0 excluded · 4 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00100	0 covered · 0 excluded · 2 gaps	p1: Alternate extracted slide text page 1; text differs from or is absent in the .slides.md page;... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human...
SRC00102	0 covered · 0 excluded · 1 gaps	p1: Extracted slide page 1 from...

Source	Unit counts	Example decisions
SRC00104	0 covered · 0 excluded · 3 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00103	0 covered · 0 excluded · 1 gaps	p1: Alternate extracted slide text page 1; text differs from or is absent in the .slides.md page;...
SRC00105	0 covered · 0 excluded · 4 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00107	0 covered · 0 excluded · 6 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00106	0 covered · 0 excluded · 4 gaps	p1: Alternate extracted slide text page 1; text differs from or is absent in the .slides.md page;... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00108	0 covered · 0 excluded · 5 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...

Source	Unit counts	Example decisions
SRC00110	0 covered · 0 excluded · 7 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00109	0 covered · 0 excluded · 5 gaps	p1: Alternate extracted slide text page 1; text differs from or is absent in the .slides.md page;... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00111	0 covered · 0 excluded · 3 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00113	0 covered · 0 excluded · 7 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...

Source	Unit counts	Example decisions
SRC00112	0 covered · 0 excluded · 3 gaps	p1: Alternate extracted slide text page 1; text differs from or is absent in the .slides.md page;... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...

C Revision and retrieval practice